Constructing Representative Microdata with Sub-National Weights: A Two-Stage Approach Combining Machine Learning Imputation and Sparse Optimization*

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Abstract

Sub-national policy analysis needs microdata that no single survey provides: joint distributions rich enough to simulate tax and transfer rules, totals that reproduce administrative aggregates across nested geographies, and a file small enough to run inside a microsimulation model. We present a two-stage pipeline that constructs such data, as implemented in POPULACE, POLICYENGINE's open-source population data stack. Stage one attaches unobserved variables to survey records with a weighted, regime-gated, sequentially chained quantile-regression-forest estimator whose interface makes an unweighted fit impossible to request by accident. Evaluated in a *population-view harness* — one latent population observed through survey views, scored by strictly proper joint distances, reweighting-invariant coverage, classifier two-sample tests, and an uncapped tail block — weighting where the design is informative dominates: the identical estimator fit unweighted is six times worse on net-worth marginal fit, and its imputed 99th percentile is 1.9 to 2.3 times the holdout's, an inflation that joint-geometry metrics tie on and only the tail block detects. Regime gating carries zero-inflated components, and sequential chaining binds at scale, where fitting

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balance-sheet components independently costs 5.8 points of two-sample AUC. Stage two reduces the enhanced candidate file to a simulable support by treating reduction as a sampling problem: L_0-regularized selection with Hard Concrete gates jointly chooses which records to keep and calibrates their weights against 32,633 administrative targets — including 24,340 congressional-district targets — from 337,704 candidate households. Selection is target-informed rather than random, and the operative finding is that gates select support while publication weights should be refit after selection: the refit 57,240-record file scores a 4.74% capped calibration loss against 5.07% for dense no-L_0 calibration and 7.55% for random subsampling. The pipeline is deployed: the sparse support produced by stage two ships as the certified Populace United States release, consumed by a companion incidence analysis of the One Big Beautiful Bill Act at national, state, and congressional-district level.

Keywords: microsimulation, imputation, survey calibration, L_0 regularization, sub-national estimation, quantile regression forests

1 Introduction

Distributional analysis of national legislation increasingly demands answers at geographies the underlying surveys were never designed to support. A bill's incidence varies with joint characteristics — wealth and program participation, tipped occupation and family structure, itemization and state of residence — that no single public-use survey observes together, and legislators consume results at the grain of states and congressional districts, where survey samples thin to a few hundred households. The microdata problem is therefore twofold. First, *fill*: attach the variables a policy model needs but the base survey lacks, from donor surveys and administrative sources, without distorting the joints that make simulation meaningful. Second, *represent*: weight the enhanced file so it reproduces administrative aggregates across nested geographies — national, state, and district — while remaining small enough to simulate interactively.

This paper presents the two-stage pipeline that answers both demands in Populace, PolicyEngine's open-source population data stack (Figure 1). Stage one is a weighted, regime-gated, sequentially chained quantile-regression-forest imputation operator. Stage two is a target-informed record-selection and calibration step: L_0-regularized gates, in the Hard Concrete relaxation, jointly choose which enhanced records to retain and assign each a positive weight, driven by tens of thousands of administrative calibration targets rather than by random subsampling.

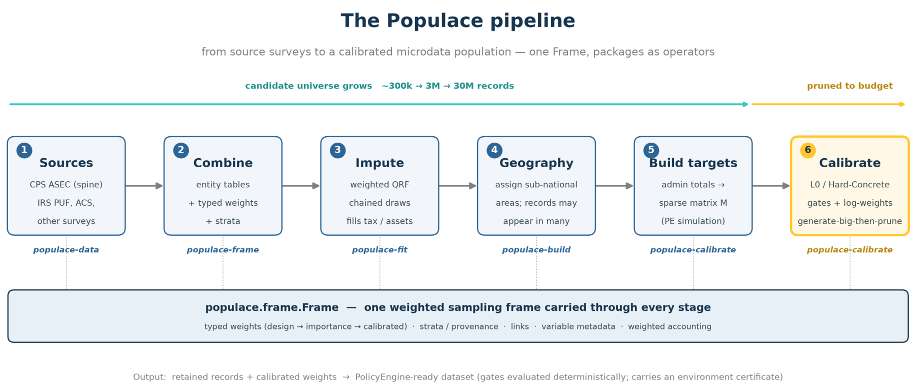

Figure 1: The two-stage pipeline. Survey and administrative sources feed a weighted, gated, chained imputation stage; the enhanced candidate file then passes through target-informed L_0 selection and calibration to produce a sparse, simulable, geography-consistent release.

Three design commitments run through both stages. *Respect the design*: imputation models fit to survey records while ignoring their weights bias imputed conditionals toward the sample rather than the population, and the damage concentrates exactly where policy analysis is most sensitive — the tails. *Make evaluation match the estimand*: we score candidate files in a population-view harness — one latent population observed through survey views, with strictly proper joint scores, a coverage axis invariant to candidate reweighting, classifier two-sample tests, and an uncapped tail block — after showing that headline joint-geometry metrics certify files whose top percentile is wrong by a factor of two. *Let the targets choose the records*: reduction to a simulable support is itself an estimation step, and selection informed by the calibration targets dominates random subsampling at equal record count.

The empirical sections establish, for stage one, that weighting where the survey design is informative is first-order (the identical estimator fit unweighted is six times worse on within-donor net-worth fit, with an imputed 99th percentile 1.9 to 2.3 times the holdout's); that regime gating carries zero-inflated income components; and that sequential chaining binds at scale, where independently fitted balance-sheet components cost 5.8 points of two-sample AUC against a classifier that detects balance sheets failing to add up. For stage two, on the full United States target surface — 32,633 materialized targets, including 24,340 congressional-district targets, over 337,704 candidate households — gates select a 57,240-record support whose post-selection refit reaches a 4.74% capped calibration loss, against 5.07% for dense calibration of the full candidate file and 7.55% for random subsampling refit the same way. The operative recipe is specific: use the gates to choose the support, then refit publication weights on it.

The pipeline is not a proposal but a deployed system. The 57,240-record support produced by stage two ships as the certified POPULACE United States release, distributed with pinned model and data versions through `policyengine.py`; a companion paper (Ghenis et al., 2026) consumes it for provision-level incidence analysis of the One Big Beautiful Bill Act at national, state, and congressional-district level, and section 11 reports what that deployment reveals about the value of target-informed selection — including a direct comparison against assignment-only district weights whose tails collapse into artifacts.

The paper proceeds in two parts. Sections 2–5 present the imputation stage: background, experimental design, the estimator, and results under the population-view harness. Sections 6–9 present the selection-and-calibration stage: background, the target surface, the L_0 method, and results. section 11 reports the deployed application, and section 12 and section 13 discuss limitations and conclude.

2 Background: imputation for microsimulation

2.1 Imputation methods in survey practice

Survey agencies and microsimulation teams draw on a small set of imputation families. *Hot-deck and statistical matching* methods donate observed values from a matched donor record; they preserve marginal distributions by construction because every imputed value is a real donor value (Andridge and Little, 2010; D'Orazio et al., 2006), and remain the standard data-fusion tool in European microsimulation practice (Sutherland and Figari, 2013). *Regression-based* methods impute from a fitted conditional model — ordinary least squares with residual draws (Von Hippel, 2007), linear quantile regression (Koenker and Bassett, 1978; Koenker, 2005) — and extend to multiple imputation for uncertainty propagation (Rubin, 1987; Kennickell, 1998). *Tree-ensemble* methods, in particular quantile regression forests (Meinshausen, 2006; Breiman, 2001), estimate full conditional distributions nonparametrically and have become the default in POLICYENGINE's data pipelines (Ghenis, 2018; Woodruff and Ghenis, 2024b) for their ability to capture nonlinear structure without per-variable specification. The `microimpute` package (PolicyEngine, 2025) implements these families under one interface and selects among them by cross-validation; its central finding — no family dominates across datasets — motivates empirical comparison on each new task rather than a prescribed method.

2.2 Chained imputation and joint structure

Imputing several variables one at a time, each conditional only on shared predictors, destroys the dependence among imputed variables; sequential (chained-equations) imputation conditions each variable on those already imputed, retaining joint structure (Raghunathan et al., 2003; Van Buuren and Groothuis-Oudshoorn, 2011). In data fusion the same issue appears as the conditional independence assumption of statistical matching: matched files preserve marginals but attenuate cross-source correlations (D'Orazio et al., 2006; Meinfelder, 2011). Synthetic-data generation faces the identical trade-off at whole-file scale (Rubin, 1993).

2.3 Weights in imputation

Design weights enter imputation practice unevenly. The model-based literature often treats weights as a nuisance — if the model is correctly specified and the design variables are conditioned on, weighting is unnecessary — while the design-based tradition weights every estimator (Little and Rubin, 2002). In production microdata pipelines the model is never correctly specified and the design variables are never fully available, so the choice is consequential: an unweighted fit targets the sample conditional, and any correlation between weights and outcomes within predictor cells propagates into the imputed population distribution. Tree ensembles add a mechanical subtlety: a fully-grown forest's *predictive distribution* is a leaf-membership distribution, which per-record `sample_weight` arguments do not reweight, so honoring weights requires materializing them into the training data — the weighted bootstrap of Section 3. Zero-inflated targets add a second subtlety: weighting the zero/nonzero gate by resampling can delete a rare sign class outright, so the gate must be weighted directly. We are not aware of prior survey-imputation work that combines weight-aware forest training, sign-regime gating, and chaining in one estimator; documenting and testing that combination is the gap this paper fills.

2.4 Evaluating imputed files

Because an imputation is a draw from an estimated conditional distribution, pointwise accuracy metrics understate quality; the relevant question is distributional. We follow the quantile-loss tradition for conditional calibration (Koenker, 2005; Meinshausen, 2006) and use Wasserstein distance for marginal fit. For the joint, the evaluation perspective this paper formalizes — one latent population, surveys as partial views of it, candidates scored through each view against held-out survey samples — has ancestry in several literatures that have not, to our knowledge, been assembled into an operational benchmark: the survey-statistics treatment of multiple sources observing one population (Lohr and Raghunathan, 2017), the missing-data view of fusion as block missingness on a single file (Little and Rubin, 2002; Rässler, 2002), strictly proper scoring of distributional claims against realized samples (Gneiting and Raftery, 2007; Székely and Rizzo, 2013), and the precision–recall–density–coverage geometry of generative-model evaluation (Naeem et al., 2020). Population-synthesis practice validates against input marginals rather than held-out views, and reviews of the field note the absence of a standard benchmark. Section 3.3 operationalizes the combination; the reweighting fragility diagnostic of Section 3.4 is, to our knowledge, new.

3 Data and experimental design for the imputation stage

Four task families span the settings the estimator is used in. Where a task imputes within one survey, the protocol of Section 4.5 applies: 80/20 donor/receiver holdouts repeated over ten paired seeds. The population-view experiment additionally projects candidates onto a second survey. Every table in Section 5 regenerates from the repository's committed run configurations; row caps and skipped cells are recorded in the run manifests.

3.1 Task 1: wealth within the SCF

The Survey of Consumer Finances (Board of Governors of the Federal Reserve System, 2023) observes household wealth in detail. From the 2022 summary extract (first implicate, survey weights) we impute `debt` — zero-inflated and nonnegative — and `networth` — sign-mixed, with a heavy right tail — from age, sex, education class, marital status, children, total income, and wage income. The two targets are chosen deliberately: their sign structures exercise the regime gates, and their strong mutual dependence exercises chaining.

3.2 Task 2: the SCF→CPS population view

Each method fits wealth conditionals on an SCF donor split and imputes them onto real CPS ASEC households sharing the predictor set — households built from the Census Bureau's ASEC public-use files (U.S. Census Bureau, 2025), read directly from `census.gov` so that no task input flows through any PolicyEngine processing. The experiment runs in two profiles. The *minimal* profile — six shared predictors, the two wealth targets of Section 3.1, a single receiver vintage (ASEC 2025) — is the controlled instantiation in which every mechanism is legible. The *populace-scale* profile mirrors the production stack's shape: the shared predictor set widens to everything the two surveys genuinely share (ten variables, adding education class, race, homeownership, and labor-force status, each mapping verified against both files' codebooks), the receiver pools three ASEC vintages (2023–2025, weights scaled to one household population, 168,852 households) as the production support spine does, and the targets widen to a four-component chained balance sheet — financial assets, nonfinancial assets, debt, then net worth — whose accounting identity networth $=$ fin $+$ nfin $-$ debt holds exactly in the extract, so chaining's contribution is directly interpretable. The candidate population — CPS demographics plus imputed wealth, under CPS household weights — is then scored through the harness of Section 4.3 against the *held-out* SCF: the joint of shared predictors and wealth, under SCF weights. The `scf_sample_reference` rows score the donor split itself against the holdout, anchoring the sampling-noise floor. This is the paper's direct test of whether conditional-distribution methods preserve joint structure that matching's conditional-independence gluing loses (Rässler, 2002; D'Orazio et al., 2006), measured where it matters: on a receiver file the donor survey never saw. Because the demographic block of every candidate is the same real CPS sample, differences between methods isolate the wealth block and its coupling to demographics; the shared demographic offset between the two surveys is common to all methods and bounded below by the reference floor.

3.3 Task 3: zero-inflated income components in the CPS

From the Census Bureau's ASEC 2025 public-use person file (income reference year 2024), read directly from `census.gov` — never a processed or enhanced artifact, which would embed the production imputations this paper is about — we impute interest and dividend income for adults from age, sex, and employment income, under the ASEC person supplement weights. (Survey files publish per-person design weights; POPULACE's production convention deliberately carries calibrated *household* weights onto persons, since its calibration operates at household grain. The task fixes the survey's own published weights, and because every method shares whatever weights the task fixes, the method comparison does not turn on this choice.) Both components are zero for most adults (41.7 and 85.2 percent, respectively) with heavy positive tails — the regime surface where gating and weighting matter most. Public-use ASEC amounts are topcoded, which truncates the extreme tail relative to an administrative or SCF-style source; the fragility comparisons of Section 5.2 are therefore conservative. This task also carries the reweighting stress test of Section 4.4: prior

POLICYENGINE production experience motivates it, where an unweighted fit broadcast rare high-value donor records across a receiver file as near-zero-weight point masses that later reweighting amplified. All fragility numbers reported here are measured under this paper's protocol.

3.4 Task 4: cross-dataset transfer

To test behavior beyond economic microdata, we run the method surface over the six OpenML AutoML Benchmark regression datasets used in the `microimpute` manuscript's appendix (Vanschoren et al., 2014) — `space_ga`, `elevators`, `brazilian_houses`, `onlinenewspopularity`, `abalone`, and `house_sales` — treating each dataset's target as the imputed column under uniform weights.

Full multi-view synthesis — one candidate scored simultaneously against CPS, SIPP (U.S. Census Bureau, 2023), and PSID (Institute for Social Research, University of Michigan, 2023) views — is the production release-gate use of the harness and is left to future work; PSID access requires registration, and the two-view experiment of Section 3.2 already identifies the joint-structure differences the paper is about.

4 Stage one: weighted, regime-gated, chained imputation

4.1 The estimator

`populace-fit` models $P(\text{targets} \mid \text{predictors})$ with a quantile-regression-forest estimator (Meinshausen, 2006; Zillow Group, 2024) wrapped in three mechanisms.

Weighted bootstrap. Random forests cannot honor a per-record `sample_weight` in their predictive distribution: a fully grown leaf holds individual training rows, and weighting the impurity criterion does not change which values a quantile query reads out. `populace-fit` therefore materializes weights into the data: before each forest is grown, training rows are resampled with replacement with probability proportional to weight, so leaf distributions reflect the weighted population. Draws from the fitted forest then sample the *weighted* conditional.

Regime gates. A numeric target's sign support — which of {negative, zero, positive} occur — defines its regime. A single regressor over a zero-inflated or sign-mixed target either drops a tail or interpolates across the empty gap at zero; hurdle-style two-part models are the classical remedy (Mullahy, 1986; Lambert, 1992). `populace-fit` fits a classifier that gates each row into a sign class and a separate magnitude forest within each nonzero sign. Regime detection is structural (it reads the unweighted support, since which signs *exist* is a fact about the variable, not the weighting), while the gate itself is weighted directly through its `sample_weight` — not by resampling, which would delete a rare sign class whose total weight share is negligible. Draws sample the gate's predicted class probabilities rather than taking the modal class, preserving the zero mass.

Sequential chaining. Targets are imputed in order, each conditioning on the predictors plus the targets already drawn (Raghunathan et al., 2003; Van Buuren and Groothuis-Oudshoorn, 2011), so dependence among imputed variables survives.

Interface contract. Fitting over a POPULACE frame reads the frame's typed survey weights by default; fitting over a plain data frame — the standalone path — requires the weights explicitly (a weight column name, a weight vector, or the literal `"none"`), and omitting them is an error rather than a silent unweighted fit. The contract makes the weight-blindness failure mode of Section 2.3 unrepresentable by accident.

4.2 Ablations and baselines

Each design choice is evaluated by knocking it out with everything else held fixed: the full estimator versus itself with `weights="none"` (the weighted bootstrap's contribution), versus fitting each target independently (chaining's contribution), and versus a single ungated forest at matched hyperparameters (gating's contribution). Baselines are standard practice: quantile-regression-forest, ordinary-least-squares, and linear quantile-regression imputers from `microimpute` (PolicyEngine, 2025), and nearest-neighbour-distance hot-deck statistical matching from `py-statmatch` (D'Orazio et al., 2021). The candidate and the plain-forest baselines share the same underlying forest implementation (Zillow Group, 2024), so their differences isolate the design choices rather than implementation quality.

4.3 The population-view harness

Per-variable metrics cannot adjudicate the question this paper cares about, because the methods differ most in how they treat the *joint*. We therefore evaluate within a framework we call the population-view harness. One latent population distribution produces individuals; each survey s is a *view* V_s of it — a variable subset, a sampling design, a measurement idiom — and the survey's holdout is a weighted sample from $V_s(P)$ (Lohr and Raghunathan, 2017). A candidate weighted file Q (any generator's) is scored by projecting it through each view and comparing $V_s(Q)$ to that view's holdout *in the view's own variable space*. Cross-survey consistency is never required — the candidate is only asked to explain each view in its own idiom — which keeps the harness usable even though surveys disagree with one another about the population they share.

Each view carries four complementary blocks. The *weighted energy distance* (Székely and Rizzo, 2013) is the sample form of a strictly proper scoring rule (Gneiting and Raftery, 2007): the true distribution uniquely minimizes its expectation, so a candidate hedged toward modal households scores strictly worse than one matching the full distribution. PRDC *coverage* (Naeem et al., 2020) — the weighted fraction of holdout points with a candidate point inside their k-nearest-neighbour radius — is the explicit anti-collapse axis; because it depends on the candidate only through its support, it is invariant to any reweighting of the candidate, making it the calibration-blind block of the scorecard. A *weighted classifier two-sample test* (cross-validated AUC of a gradient-boosted classifier at equal class mass; 0.5 is indistinguishable) is the omnibus check. The fourth block exists because we caught the first three missing a real failure: pairwise sample-geometry metrics run on capped subsamples, and a candidate whose imputed wealth 99th percentile is double the holdout's can tie on energy distance because the discrepant mass is a sliver of standardized pairwise space. Economic variables live in heavy right tails, so each *imputed* column also carries a *tail block*, computed on the full weighted samples with no cap: its weighted Wasserstein-1 distance to the holdout (scaled by the holdout's weighted standard deviation) and its weighted q90 and q99 ratios (candidate over holdout; 1 is a perfect tail). Holdouts are used nowhere upstream, so the harness is a non-self-referential test surface; a sampling-noise floor — the donor split itself scored as a candidate — anchors what "as good as another sample of the survey" means on each axis, including how much a tail ratio wobbles under sampling noise alone.

4.4 Marginal and stress metrics

All metrics are computed under survey weights, on held-out receiver rows. *Weighted pinball loss*, averaged over a decile grid, scores conditional distributional calibration (Koenker, 2005). *Weighted Wasserstein-1 distance* to the weighted donor distribution scores marginal fit. *Zero-share error* scores preservation of the exact-zero mass of zero-inflated targets. Fi-

Method	Debt pinball	Debt W_1	Debt zero-share err.	Net worth pinball	Net worth W_1
populace-fit	49,201 (2,637)	**13,001** (6,285)	0.020 (0.012)	551,317 (36,012)	**135,871** (51,106)
– unweighted	50,218 (2,452)	32,247 (16,930)	**0.012** (0.014)	703,457 (77,173)	829,158 (445,108)
– unchained	49,201 (2,637)	**13,001** (6,285)	0.020 (0.012)	556,622 (23,349)	158,606 (44,310)
– ungated/unchained forest	50,433 (3,120)	16,619 (5,358)	0.018 (0.018)	560,819 (36,903)	234,367 (142,057)
OLS	70,053 (17,637)	156,295 (74,822)	0.226 (0.018)	1,116,572 (383,007)	3,565,923 (2,062,063)
Quantile regression	49,797 (3,391)	29,460 (20,456)	0.151 (0.031)	534,420 (37,801)	295,470 (110,986)
NND hot deck (py-statmatch)	49,537 (2,627)	13,770 (5,141)	0.017 (0.013)	644,921 (51,255)	509,013 (399,757)
Weighted marginal draw	**48,277** (2,586)	14,617 (4,732)	0.020 (0.017)	**443,554** (20,651)	215,398 (87,264)

Table 1: Wealth imputation within the SCF (debt and net worth from demographics and income): weighted pinball loss, Wasserstein-1 to the weighted donor distribution, and zero-share error, mean (sd) over ten paired seeds. Bold marks the best non-reference value per column. Lower is better.

nally, *reweighting fragility* formalizes the landmine diagnostic: over the family of bounded multiplicative reweightings $w_i \mapsto m_i w_i$ with $m_i \in [1/\kappa, \kappa]$ (the hard weight-ratio bounds production calibration guards use), the worst-case single-record share of a target aggregate has the closed form $\kappa^2 c^*/(\kappa^2 c^* + S - c^*)$, where c^* is the largest baseline contribution $w_i|a_i|$ and S their sum. We report it at $\kappa = 5$ for every method next to the held-out truth's own fragility — the exposure a method should not exceed.

4.5 Protocol

Every task uses an 80/20 donor/receiver holdout repeated over ten seeds, with splits paired across methods: the split is a pure function of the seed, so every method fits and is scored on identical partitions. Task inputs are frozen once and pinned by hash; only the method varies within a sweep. All results tables are generated from the sweep artifacts by the repository's command line; no result is hand-entered. Where a method is unavailable or a metric undefined for a task, the cell is reported as absent rather than omitted.

5 Imputation results

Throughout, `plain_qrf` — the ungated, unchained forest — is the same estimator and configuration as the `microimpute` QRF baseline, so it appears once, as an ablation row; OLS, quantile regression, and hot deck are the remaining standard-practice baselines, and the weighted marginal draw is the unconditional lower bound.

Method	Interest pinball	Interest zero-share	Dividend pinball	Dividend zero-share
populace-fit	1,491 (63.58)	0.013 (0.011)	320.68 (45.26)	0.006 (0.004)
– unweighted	**1,491** (63.62)	0.011 (0.007)	320.68 (45.29)	0.008 (0.005)
– unchained	1,491 (63.58)	0.013 (0.011)	320.69 (45.26)	**0.006** (0.005)
– ungated/unchained forest	1,493 (63.34)	0.014 (0.011)	322.17 (45.24)	0.044 (0.010)
OLS	2,527 (63.57)	0.422 (0.012)	961.52 (112.29)	0.844 (0.005)
Quantile regression	1,493 (63.23)	0.270 (0.016)	320.91 (45.19)	0.251 (0.011)
NND hot deck (py-statmatch)	1,493 (63.89)	0.021 (0.019)	320.94 (45.22)	0.009 (0.009)
Weighted marginal draw	1,491 (63.61)	**0.011** (0.013)	**320.68** (45.25)	0.006 (0.004)

Table 2: Zero-inflated income components in the raw CPS (interest and dividend income for adults): weighted pinball loss and zero-share error, mean (sd) over ten paired seeds.

5.1 Wealth within the SCF

Table 1 reports the within-SCF task. The headline is the weighting ablation: switching the candidate to `weights="none"` degrades net-worth Wasserstein-1 by a factor of six (135,871 to 829,158) and debt by a factor of 2.5 (13,001 to 32,247). The SCF's list-sample design oversamples wealthy households at low weight, so an unweighted fit learns the sample's wealth distribution rather than the population's — this is the weight-blindness failure mode of Section 1 measured directly, and no other design choice comes close to its effect size. Gating and chaining contribute more modestly: the ungated, unchained forest sits at 234,367 on net-worth W_1 (1.7 times the candidate), and removing chaining raises net-worth W_1 by 17 percent (135,871 to 158,606).

Against the baselines, the candidate's margins are largest where tails and joints matter: hot-deck matching nearly ties it on debt marginal fit (13,770 vs. 13,001) but is 3.7 times worse on net worth (509,013), OLS is catastrophic on both (156,295 and 3.6 million — normal residuals cannot represent a wealth distribution), and linear quantile regression leaves 15 percent of the debt zero mass unaccounted for, against at most 2 percent for the forest-based methods.

One result reads wrong until the metric is examined: the *unconditional* weighted marginal draw posts the best pinball loss on both targets. With predictors as weak as demographics and income are for wealth, receiver-side marginal metrics cannot distinguish a method that draws from the correct weighted marginal while ignoring predictors entirely from one that also gets the conditionals right. This is not an artifact to explain away; it is the motivating observation for the population-view harness of Section 5.5, which scores the joint.

Method	Interest fragility	(truth)	Dividend fragility	(truth)
populace-fit	0.384 (0.051)	0.372 (0.043)	0.675 (0.100)	0.614 (0.104)
– unweighted	0.395 (0.043)	0.372 (0.043)	0.699 (0.057)	0.614 (0.104)
– unchained	0.384 (0.051)	0.372 (0.043)	0.661 (0.100)	0.614 (0.104)
– ungated/unchained forest	0.364 (0.064)	0.372 (0.043)	**0.633** (0.087)	0.614 (0.104)
OLS	0.055 (0.010)	0.372 (0.043)	0.055 (0.009)	0.614 (0.104)
Quantile regression	**0.368** (0.105)	0.372 (0.043)	0.444 (0.073)	0.614 (0.104)
NND hot deck (py-statmatch)	0.391 (0.027)	0.372 (0.043)	0.647 (0.140)	0.614 (0.104)
Weighted marginal draw	0.356 (0.042)	0.372 (0.043)	0.746 (0.067)	0.614 (0.104)

Table 3: Reweighting fragility at $\kappa = 5$: worst-case single-record share of each component's aggregate over bounded multiplicative reweightings, next to the held-out truth's own fragility. The target is the truth's exposure — bold marks the closest method; substantially lower means the method fails to generate realistic tails, substantially higher means it plants landmines.

5.2 Zero-inflated components and reweighting fragility

Table 2 reports the CPS components task. Pinball loss barely separates the non-OLS methods — age, sex, and earnings carry little conditional signal for capital income — but the zero mass separates them sharply. The candidate holds zero-share error to 0.6 and 1.3 percentage points (dividends, interest); the ungated forest drifts to 4.4 points on dividends; linear quantile regression misses by 25–27 points because a 99-level linear quantile grid cannot hold a five-sixths zero mass; and OLS misses the dividend zero share by 84 points — with normal residual draws, nearly every adult becomes a dividend recipient. Hot deck, which donates observed values, holds the zero mass within 0.9–2.1 points, as expected.

Table 3 reports the landmine diagnostic. Two readings matter. First, the held-out truth is itself concentrated: under $\kappa = 5$ reweighting a single record can carry 37 percent of the interest aggregate and 61 percent of the dividend aggregate at this sample size — heavy-tailed capital income is genuinely fragile, and a faithful method should *match* that exposure, not minimize it. Second, every forest- and donation-based method tracks the truth's fragility closely, while OLS sits far below it (0.06 against 0.37 and 0.61 — understating the true exposure by factors of roughly 7 and 11): its aggregate is spread across many moderate records because the method never generates realistic tails — the same failure that its zero-share and Wasserstein numbers show from other angles. No method exceeds the truth's exposure materially on this within-survey task; the landmine pathology of Section 1 arises when unweighted cross-file fits meet downstream reweighting, and the direct evidence for it here is the tail block of the population-view experiment (Section 5.5).

Task	Target	Metric	Δ unweighted	Δ unchained	Δ plain forest
cps components	dividend_income	pinball_loss	0.000	0.005	1.48
cps components	dividend_income	wasserstein1	36.11	11.29	308.65
cps components	dividend_income	zero_share_error	0.002	-0.000	0.038
cps components	interest_income	pinball_loss	-0.016	0.000	1.57
cps components	interest_income	wasserstein1	-26.28	0.000	116.69
cps components	interest_income	zero_share_error	-0.002	0.000	0.001
scf wealth	debt	pinball_loss	1,017	0.000	1,232
scf wealth	debt	wasserstein1	19,246	0.000	3,618
scf wealth	debt	zero_share_error	-0.008	0.000	-0.001
scf wealth	networth	pinball_loss	152,140	5,305	9,502
scf wealth	networth	wasserstein1	693,287	22,735	98,496
scf wealth	networth	zero_share_error	-0.000	0.000	0.001

Table 4: Paired-seed deltas, ablation minus candidate (positive = ablation worse on these lower-is-better metrics), mean over ten seeds.

Method	abalone	brazilian_houses	elevators	house_sales	onlinenewspopularity	space_ga	Mean rank
populace-fit	0.254	166.64	0.000	7,080	274.67	0.008	3.83
– unweighted	0.257	184.92	0.000	6,744	233.10	0.008	**3.33**
– unchained	0.254	166.64	0.000	7,080	274.67	0.008	3.83
– ungated/unchained forest	0.174	191.36	0.000	8,461	546.16	0.012	5.00
OLS	0.532	212.09	0.002	90,518	6,117	0.015	8.83
Quantile regression	0.321	212.04	0.001	24,115	382.98	0.085	7.83
NND hot deck (py-statmatch)	0.176	199.12	0.000	10,212	161.43	0.010	4.00
Weighted marginal draw	0.148	210.50	0.000	7,135	143.52	0.011	**3.33**

Table 5: Wasserstein-1 to the donor distribution across the six OpenML regression datasets (uniform weights), mean over ten paired seeds, with mean rank across datasets.

5.3 Ablations: attributing the gains

Table 4 attributes the candidate's behavior to its design choices with paired-seed deltas. The ordering is task-dependent in an informative way. On the SCF — informative weights, heavy tails — weighting dominates (+693,287 net-worth W_1 when removed), the structural forest machinery matters next (+98,496), and chaining contributes a consistent but smaller +22,735. On the CPS components — weakly informative weights, extreme zero inflation — weighting is approximately free ($\Delta \approx 0$), while gating carries the load (+534 dividend W_1 for the plain forest, and the zero-share drift of Table 2). The design choices are complements across regimes, not redundant safeguards: which one binds depends on the survey's design and the target's structure, and none of them hurts where it does not help.

5.4 Cross-dataset transfer

Table 5 runs the surface over the six OpenML regression datasets under uniform weights. Two observations. First, with weights uninformative by construction, the candidate and its

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.009 (0.006)	0.987 (0.009)	0.496 (0.023)	0.013 (0.003)	1.10 (0.202)
populace-fit	**0.139** (0.026)	0.960 (0.014)	0.732 (0.013)	0.031 (0.007)	0.696 (0.112)
– unweighted	0.140 (0.026)	0.958 (0.014)	**0.731** (0.016)	0.159 (0.041)	1.93 (0.255)
– unchained	0.139 (0.026)	0.958 (0.016)	0.742 (0.013)	0.029 (0.009)	0.765 (0.177)
– ungated/unchained forest	0.140 (0.027)	0.957 (0.012)	0.775 (0.008)	0.029 (0.007)	0.805 (0.156)
OLS	0.166 (0.045)	0.732 (0.087)	0.988 (0.005)	0.434 (0.282)	**0.967** (0.505)
Quantile regression	0.141 (0.027)	0.938 (0.028)	0.927 (0.018)	0.052 (0.013)	0.530 (0.118)
NND hot deck (py-statmatch)	0.141 (0.027)	**0.962** (0.013)	0.753 (0.014)	0.051 (0.016)	0.762 (0.148)
Weighted marginal draw	0.139 (0.026)	0.937 (0.005)	0.848 (0.008)	**0.014** (0.004)	1.09 (0.222)

Table 6: The population-view experiment, *minimal* profile (six shared predictors, two targets, ASEC 2025 receiver): candidates assembled from real CPS demographics plus each method's imputed wealth, under CPS household weights, scored against the held-out SCF through the SCF view. The first row is the sampling-noise floor. Energy distance lower is better; coverage higher; C2ST AUC of 0.5 is indistinguishable; net-worth W_1/sd lower is better; a q99 ratio of 1 is a perfect extreme tail.

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.012 (0.005)	0.986 (0.003)	0.499 (0.021)	0.013 (0.003)	1.10 (0.202)
populace-fit	**0.118** (0.024)	0.969 (0.009)	0.761 (0.026)	0.027 (0.008)	0.767 (0.190)
– unweighted	0.120 (0.024)	**0.971** (0.008)	**0.754** (0.011)	0.133 (0.038)	2.25 (0.443)
– unchained	0.118 (0.024)	0.967 (0.011)	0.819 (0.015)	0.028 (0.006)	0.699 (0.112)
– ungated/unchained forest	0.126 (0.025)	0.965 (0.011)	0.817 (0.009)	0.033 (0.022)	1.15 (0.445)
OLS	0.152 (0.059)	0.877 (0.055)	0.995 (0.002)	0.435 (0.282)	**0.963** (0.503)
Quantile regression	0.120 (0.026)	0.951 (0.011)	0.970 (0.007)	0.051 (0.013)	0.485 (0.098)
NND hot deck (py-statmatch)	0.119 (0.025)	0.971 (0.010)	0.768 (0.009)	0.024 (0.008)	0.747 (0.107)
Weighted marginal draw	0.119 (0.024)	0.931 (0.015)	0.968 (0.003)	**0.015** (0.003)	1.12 (0.241)

Table 7: The population-view experiment, *populace-scale* profile (ten shared predictors, four chain-ordered balance-sheet targets, pooled ASEC 2023–2025 receiver). Columns as in Table 6.

unweighted ablation are statistically the same method, and they rank near the top (mean ranks 3.8 and 3.3) alongside the unconditional marginal draw (3.3); the linear methods rank last (7.8 and 8.8). Second — and more important for this paper's argument — the best mean rank on a *marginal* distance is achieved by a method with no conditional structure at all. Marginal metrics saturate: they reward drawing from the right marginal and cannot rank what imputation is actually for. The population-view harness exists because of exactly this ceiling.

5.5 The SCF→CPS population view

Tables 6 and 7 report the population-view experiment in its two profiles. The floor behaves as designed in both: another sample of the same survey is statistically indistinguishable (energy 0.009 and 0.012, coverage 0.987 and 0.986, AUC 0.496 and 0.499), and its q99 ratio of 1.10 calibrates how much a 99th-percentile ratio moves under sampling noise alone. Four findings follow.

First, *the weighting failure is a tail failure, and only the tail block sees it*. The unweighted ablation ties the candidate on energy distance, coverage, and the C2ST in both profiles — and its imputed net-worth q99 is 1.93 times the holdout's in the minimal profile and 2.25 times at populace scale, with tail W_1/sd five times the candidate's (0.159 vs. 0.031 minimal; 0.133 vs. 0.027 at scale). Fitting the SCF unweighted learns the wealthy oversample's conditionals, and the resulting inflation concentrates beyond the resolution of capped pairwise geometry: we report this as a finding about evaluation practice. Any harness whose joint metrics subsample — which at these sample sizes they must — needs an uncapped tail block for heavy-tailed economic variables, or it will certify files whose top percentile is wrong by a factor of two.

Second, *chaining is a joint property, and it binds at scale*. In the minimal profile (two targets) the unchained ablation is indistinguishable from the candidate. At populace scale, where the four targets satisfy the accounting identity networth = fin + nfin − debt exactly in the donor, fitting them independently costs 5.8 points of C2ST AUC (0.819 vs. 0.761) while leaving every marginal and tail statistic essentially unchanged — the classifier detects balance sheets that do not add up, which no marginal metric can. The same pattern holds for gating: the ungated forest loses 5.6 AUC points at scale (0.817).

Third, *conditioning richness makes the omnibus test decisive*. The unconditional weighted marginal draw — which ties or beats every method on marginal metrics throughout this paper — moves from AUC 0.848 in the minimal profile to 0.968 at populace scale, while the candidate stays closest to the floor (0.732 and 0.761). With ten shared predictors and a four-component wealth block, a classifier finds the missing demographics–wealth coupling almost perfectly. This is the empirical case for carrying strong auxiliary predictors in a production imputation stack: richer conditioning is what turns joint-structure failures from undetectable into obvious.

Fourth, *hot-deck matching is a genuine near-peer on this task*. It sits within noise of the candidate on energy and coverage in both profiles, within 2.1 and 0.7 AUC points, and its donated values slightly beat the candidate's on tail W_1 at scale (0.024 vs. 0.027) — donation preserves marginals and tails by construction. The candidate's advantages over matching live where this task does not press hard: informative-weight regimes (Table 1), extreme zero inflation (Table 2), and multi-target coherence (the chaining result above, which matching handles only through whole-record donation at the cost of reusing donors). One more tail observation cuts against every conditional method, the candidate included: forest draws do not extrapolate beyond training support, so their imputed q99 ratios sit at 0.70–0.81 — they understate the extreme tail by 20–30 percent where donation-based draws sit near 1. No method in this surface gets the extreme tail right for free; the tail block is what keeps that visible.

6 Background: calibration and record selection

6.1 Subnational microsimulation and spatial reweighting

Subnational microsimulation starts from a recurring problem: a national survey carries the behavioural and demographic detail needed for policy modelling, but it does not contain enough observations in every local area to support direct estimation. Spatial microsimulation addresses that gap either by reweighting existing survey records or by building synthetic small-area populations from aggregate constraints (Tanton, 2014; O'Donoghue et al., 2014).

This distinction matters for the present work, because our pipeline is a reweighting system built on observed CPS households. It places households across geographies to create local variants, but the core problem remains the calibration of survey-based microdata to administrative targets, which keeps calibration methods as the closest reference point.

The spatial microsimulation literature supplies further context. Williamson et al. (1998), Huang and Williamson (2001), and Harland et al. (2012) describe combinatorial and synthetic-population approaches that search over record assignments area by area. Tanton et al. (2011) and Lovelace and Dumont (2016) discuss reweighting for small-area estimation and policy analysis. These methods usually target one geographic layer at a time, whereas our setting asks one weighted dataset to reproduce state and national targets together.

6.2 Survey calibration

Let $i = 1, \ldots, n$ index sampled units, let d_i denote the initial design weight for unit i, and let $\mathbf{x}_i \in \mathbb{R}^p$ denote the auxiliary variables used for calibration. Survey calibration seeks adjusted weights w_i whose weighted totals match known population totals:

$$\sum_{i=1}^{n} w_i \mathbf{x}_i = \mathbf{T}, \tag{1}$$

where $\mathbf{T} \in \mathbb{R}^p$ is the vector of published totals (Deville and Särndal, 1992; Särndal, 2007). The auxiliary variables may be counts, indicators, or continuous quantities such as income components.

The subnational problem extends this setup in two ways. The target vector draws on several administrative sources, so it mixes count and dollar constraints. The constraints are also nested by geography: district totals should sum to state totals, which should sum to national totals after uprating and reconciliation. At production scale this yields a large target system with substantial collinearity across rows.

6.3 Calibration estimators

Two classical methods are the standard references for this kind of problem. Generalized regression (GREG) minimizes a distance from the initial weights subject to the calibration equations (Deville and Särndal, 1992; Särndal, 2007). It accepts quantitative targets directly and is well understood, but it can return negative weights and becomes harder to solve as the target system grows and its rows become collinear. Negative weights are acceptable in some estimation settings and awkward in a microdata file meant for tax-benefit simulation, where a household is expected to represent positive population mass.

Iterative proportional fitting (IPF, or raking; Deming and Stephan, 1940; Ireland and Kullback, 1968) adjusts weights multiplicatively so that selected margins match published totals. It preserves non-negativity and avoids matrix inversion, which makes it a natural choice for count-style margins. In spatial microsimulation and population synthesis, raking is commonly paired with a sampling step in one of two orders: draw a sample first and rake that fixed sample, or rake the full file first and integerise the fractional weights into a finite population. Iterative proportional updating and related extensions handle household and person margins simultaneously (Pritchard and Miller, 2012). These raking-based workflows are natural comparators for pruning methods on categorical margin surfaces. They are not direct full-surface baselines here: classical IPF is built for non-negative categorical margins, whereas the production target surface mixes counts, dollar totals, overlapping families, near-zero targets, and hierarchical constraints. Generalized raking or entropy calibration can extend the idea to broader linear constraints, but then the method becomes a general calibration optimizer rather than the classical IPF workflow.

A third approach treats calibration as an optimization problem to be solved numerically. The two methods above are special cases of one problem: Deville and Särndal (1992) define the calibrated weights as those closest to the design weights, under a chosen distance, subject to reproducing the targets, with GREG and raking the closed-form solutions for particular distances. When the target system is large, collinear, mixes count and dollar constraints, and restricts weights to be positive, the classical Newton and Lagrange solvers become hard to apply or fail to converge, which motivates minimizing the calibration objective directly. Espuny-Pujol et al. (2018) do so with a global optimization that enforces the range restrictions on the weights explicitly, and minimum-distance reweighting has long been used this way in tax microsimulation (Creedy, 2003). At the scale of an enhanced microdata file the practical solver is stochastic gradient descent: the weights are parameterized to stay positive, typically in log space; a loss measures the relative discrepancy between the weighted estimates and the targets; and an optimizer such as Adam (Kingma and Ba, 2015) minimizes it, which scales to thousands of mixed, hierarchical targets without inverting a matrix. Woodruff and Ghenis (2024a) calibrate PolicyEngine's enhanced CPS in exactly this way.

These estimators answer a calibration question rather than a sampling one: given a fixed set of records, they find weights that match the targets. The method studied here selects records and fits weights at the same time, extending the gradient-based calibration just described with explicit record selection. Classical calibration still matters for the comparison set, because it can be combined with selection: sampling followed by raking is the classical reduce-then-calibrate analogue to random sampling followed by gradient reweighting, while raking followed by integerisation is the classical calibrate-then-reduce analogue to sampling from dense calibrated weights.

6.4 Reducing microdata by selecting records

Reducing a large candidate dataset to a fixed budget is a sampling problem with a long statistical history, and several established methods reduce a microdata file by selecting actual records to match known totals. They differ mainly in where calibration enters: before selection, after selection, or inside the selection step itself. We review the broad option space because it defines what a reasonable analyst might try: random sampling followed by calibration, calibration followed by integerisation, raking-based variants of both, balanced sampling, combinatorial optimisation, and sparse optimization relaxations. Not all of these are equally applicable to the production problem. The target surface here is large, mixed, and hierarchical, so methods built for categorical margins or discrete area-by-area searches are reference points rather than direct full-surface baselines. The matched-budget comparison in Section 8 therefore focuses on the tractable full-surface set: informed L_0, informed L_0 followed by ordinary refitting on the selected records, dense no-L_0 calibration on the full candidate universe, random sampling with reweighting, and a no-refit random sample of dense calibrated weights scaled back to total mass. Raking-based reductions, survey-weight sampling, convex sparse calibration, balanced sampling, and combinatorial optimisation remain published comparators that motivate robustness checks where their assumptions fit cleanly. We also note the coreset view from machine learning, where a weighted subset is chosen so that a model fit on the subset approximates the fit on the full data (Feldman, 2020; Bachem et al., 2017).

6.4.1 Random sampling

The simplest reduction draws records at random. Under simple random sampling without replacement each record has inclusion probability n/N and design weight N/n, and the Horvitz–Thompson estimator recovers population totals without bias (Horvitz and Thompson, 1952; Cochran, 1977; Särndal et al., 1992). A random subsample preserves the conditional distribution of the data in expectation, but it does not reproduce administrative totals, so it has to be calibrated after the draw. Random selection is also the reference that the data-reduction literature treats as the genuine test: informativeness-based schemes are measured against it (Ma et al., 2015), and pruning methods in deep learning often fail to beat it at high reduction ratios (Guo et al., 2022). In our setting it is the reduce-first, calibrate-after baseline, a random draw followed by gradient-descent reweighting to the targets. Replacing the gradient reweighting step with raking gives the classical sampling-then-raking workflow; it is directly comparable on a raking-compatible categorical margin subset, but not on the full mixed count-and-dollar target surface without changing the problem into generalized calibration.

6.4.2 Raking, survey-weight sampling, and integerisation

A second approach calibrates first and reduces second. Given fractional weights from a calibration step, survey-weight sampling selects records with probability proportional to those weights, which turns a reweighted file into a finite, integer-weighted population that approximately preserves both the calibrated totals and the distribution. This is unequal-probability, probability-proportional-to-size sampling in the survey tradition (Hansen and Hurwitz, 1943; Horvitz and Thompson, 1952), applied to microsimulation as integerisation. The upstream calibration can be IPF/raking, GREG, or a numerical optimizer; the downstream question is how to convert fractional weights into a finite file. Lovelace and Ballas (2013) catalogue the common integerisation methods and show that truncate-replicate-sample reproduces the targets most accurately while fixing the reduced population to an exact size; earlier top-up schemes are due to Ballas et al. (2005). Because the selection probabilities are the calibration weights, survey-weight sampling presupposes a calibration step and is the calibrate-then-reduce member of the literature most adjacent to our setting. It is informed selection, but it prioritises records by the population mass each carries rather than by their contribution to target fit: a record is retained with probability proportional to its prior weight, so the step keeps records that represent more population, not records that help match a target the fit would otherwise miss. The administrative targets enter only through the upstream weights, and the sampling preserves that calibrated distribution instead of re-optimising against the targets. We therefore treat it as a related literature method rather than a target-fitting baseline run here, and note it as a natural calibrate-then-reduce comparator for a future sweep (Section 10). A raking-then-TRS variant is the corresponding classical baseline on a categorical margin surface, but it does not naturally extend to the full mixed production target surface without replacing IPF with a broader calibration solver.

6.4.3 Balanced sampling

Balanced sampling selects a fixed-size sample whose Horvitz–Thompson estimates reproduce auxiliary totals exactly or nearly exactly. The cube method of Deville and Tillé (2004) is the standard construction: it chooses inclusion indicators while preserving prescribed balancing equations, then uses the resulting sample weights for estimation. This makes balanced sampling a target-aware selection method rather than a post-sampling calibration method, and therefore a relevant published neighbour of L_0 selection. Its fit to the present problem is not exact. Classical balanced sampling usually treats the auxiliary totals as design constraints with predetermined inclusion probabilities, whereas the production calibration surface here mixes categorical counts, dollar totals, and hierarchical relationships and then fits positive weights after selection. Still, it is an important reference point for the idea that the sample itself, not only its weights, can be chosen to preserve known totals.

6.4.4 Convex sparse calibration

A final family keeps the continuous optimization setup but replaces the direct retained-record count with a convex sparsity surrogate. An L_1 penalty on fitted weights, implemented with proximal soft-thresholding, can drive weights exactly to zero while preserving the same calibration loss, and it is tractable on the full target surface. Its limitation is structural rather than computational: the single penalty couples how many records survive with how much the surviving weights shrink, so its tuning parameter is not a direct retained-record-count control in the way the expected L_0 norm is. We therefore read it as a penalty-based analogue of the L_0 selector that this study does not implement, and one we expect to make a weaker sampler for that reason; Section 6.6 develops the mechanism and the coupling argument.

6.4.5 Informed selection

These methods share the idea that a carefully chosen weighted subset can stand in for a larger dataset, but they reach it differently: random sampling ignores the targets and repairs the gap afterwards, survey-weight sampling inherits the targets from a prior calibration, balanced sampling tries to preserve auxiliary totals through the sample design, and convex sparse calibration remains in the same optimizer but changes the penalty. A fourth tradition makes selection itself the calibration: combinatorial optimisation chooses a combination of records, with integer multiplicities, to match the benchmark totals directly, searched with metaheuristics such as simulated annealing (Kirkpatrick et al., 1983; Williamson et al., 1998; Voas and Williamson, 2000). It is the closest precedent for target-informed selection, but it is discrete and gradient-free; we note it as related work rather than include it in the matched-budget comparison. The method studied here also pursues the targets at selection, but with a continuous, gradient-based relaxation that fits the selection and the weights jointly. The next subsection sets out that relaxation.

6.5 L_0 regularization and the Hard Concrete distribution

The L_0 norm counts the non-zero entries in a vector. Applied to record weights, it counts the records that remain active, which is exactly the size of the retained sample. Selecting a subset of a chosen size is therefore an L_0-penalized problem. The difficulty is that the L_0 norm is non-differentiable and combinatorial, so it cannot be minimized by gradient descent in its raw form.

Louizos et al. (2018) make the penalty differentiable by attaching a stochastic gate $z_i \in [0, 1]$ to each parameter and optimizing the expected L_0 norm. The gate is drawn from a Hard Concrete distribution: a binary concrete (continuous-relaxation) variable stretched beyond the unit interval and then clipped back into it. With a learned location parameter

$\log \alpha_i$ and a fixed temperature β, draw $u \sim \mathrm{Uniform}(0, 1)$ and set

$$s_i = \sigma\left(\frac{\log u - \log(1-u) + \log \alpha_i}{\beta}\right), \tag{2}$$

$$\bar{s}_i = s_i(\zeta - \gamma) + \gamma, \tag{3}$$

$$z_i = \min\bigl(1, \max(0, \bar{s}_i)\bigr), \tag{4}$$

where σ is the logistic function and $\gamma < 0 < 1 < \zeta$ are fixed stretch bounds. Stretching s_i from $(0, 1)$ to (γ, ζ) and clipping to $[0, 1]$ places point masses at exactly 0 and exactly 1, so a gate can switch a record fully off or fully on while staying differentiable in between.

Because the gate has a closed-form probability of being non-zero, the expected L_0 norm is differentiable. The probability that gate i is active is

$$\Pr(z_i \neq 0) = \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right), \tag{5}$$

and the L_0 penalty is the sum of these probabilities over all records, $\sum_i \Pr(z_i \neq 0)$, which is the expected number of retained records; minimizing it pushes gates toward zero. At test time the noise is removed and the gate is evaluated deterministically,

$$\hat{z}_i = \min(1, \max(0,\ \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma))\,, \tag{6}$$

which returns an exact zero for any record driven far enough off and an exact one for any record driven far enough on, with intermediate values in between. The exact zeros make the weight vector exactly sparse; a record left with an intermediate gate is retained with its fitted weight scaled by the gate.

The two pieces depend on each other, which is the point worth stressing for readers new to this construction. The L_0 penalty states the objective, that fewer active records are preferred, but on its own it is not optimizable by gradient methods. The Hard Concrete gate makes that objective differentiable, but on its own, without the penalty, it carries no pressure toward sparsity and leaves every gate open. Used together they turn record selection into a term the optimizer can minimize alongside calibration error. This is what makes the retained-record count a control variable rather than the result of a separate thresholding step.

6.6 Convex sparse selection (L_1)

The L_0 count is the natural sparsity objective but a hard one. The standard convex surrogate replaces it with the L_1 norm of the weights, $\|\mathbf{w}\|_1 = \sum_i |w_i|$, which is the LASSO of Tibshirani (1996). Adding the penalty to the calibration loss gives the convex program

$$\min_{\mathbf{w} \geq 0} \mathcal{L}_{\text{cal}}(\mathbf{w}) + \lambda_{L_1} \sum_{i=1}^{n} w_i, \tag{7}$$

where λ_{L_1} sets the strength of the penalty and the weights are non-negative here, so $|w_i| = w_i$. The L_1 norm is the standard tool for sparse regression and feature selection: unlike the smooth L_2 (ridge) penalty, which shrinks coefficients but rarely sets them to zero, its corner at the origin drives small coefficients to exact zeros. The program is convex and is minimized by proximal gradient descent, each gradient step followed by a soft-thresholding step $w_i \leftarrow \max(w_i - \tau, 0)$ that subtracts a fixed amount $\tau \propto \lambda_{L_1}$ from every weight and clips at zero (Beck and Teboulle, 2009), so a record whose pull on the targets does not clear the threshold is dropped while the rest are kept and shrunk. The trade-off is structural: the single penalty λ_{L_1} controls both how many records survive and how much their weights shrink, because the same norm that zeroes small weights also pulls the surviving ones toward zero, so forcing a small sample forces heavy shrinkage on the records that remain. This coupling is why we expect L_1 to be a weaker selector than L_0: the Hard Concrete gates separate the decision to keep a record from the weight it carries, whereas a single L_1 penalty cannot. We therefore do not implement L_1 in this study and treat it as an analogous penalty method worth testing directly in future work (Section 10), where running it against L_0 on the same surface would show whether the coupling actually costs it.

6.7 Microsimulation dataset engines

Beyond the calibration step itself, Woodruff and Ghenis (2024a) embed that gradient-based reweighting in a national pipeline that first imputes tax variables from the IRS Public Use File onto CPS records, and that regularizes the weights with dropout rather than an explicit selection mechanism. The present work keeps the gradient-based calibration but replaces dropout with the L_0 and Hard Concrete gates of the previous subsection, so that record selection becomes explicit and the fitted weights are exactly sparse. It also runs within POPULACE, a pipeline that expresses dataset construction as a sequence of modular stages over a shared data structure. The data sources, the imputation model, the geographic assignment, and the calibration method are interchangeable components rather than fixed steps, which lets the same pipeline support different countries and different methodological choices. Section 7 describes the pipeline and the configuration used here.

7 Data and target surface for the calibration stage

The calibration problem is defined by two inputs: the household microdata that form the candidate universe, and the administrative targets the calibrated dataset must reproduce. This section describes how POPULACE assembles the candidate microdata and how this study selects its targets. The pipeline is the harness that produces the calibration problem; it is not itself the object of evaluation.

7.1 Populace as a construction pipeline

POPULACE builds a microsimulation dataset by passing a single weighted data structure—a sampling frame holding the records, their weights, and provenance—through a sequence of stages. Each stage is a separate package that reads and writes that shared frame: it loads the source surveys, combines them into one record universe, imputes missing variables, assigns geography, constructs the calibration targets, and calibrates. Because every stage operates on the same structure, the imputation model and the calibration method are components that can be swapped without rewriting the pipeline, and the method studied here is one calibration option among those the pipeline supports.

This modular structure makes the candidate universe a property of the configuration rather than an accident of the code: the source-loading and imputation stages set what information each record carries, and calibration sets which records survive and at what weight. POPULACE follows a generate-then-prune design, assembling the candidate universe once and pruning it to a deployable budget at the calibration stage.

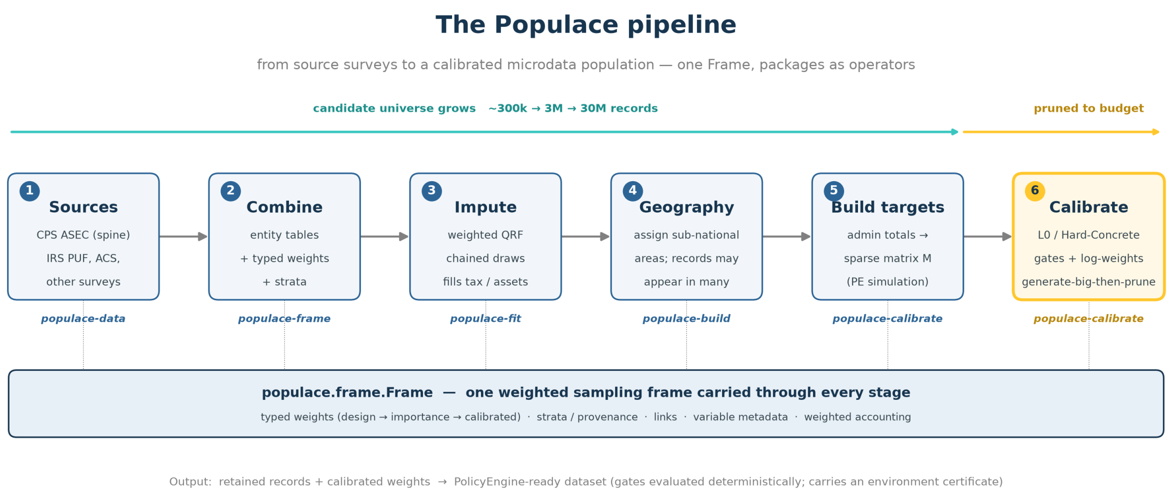

Figure 2: POPULACE pipeline for the configuration used in this paper: load and combine the source surveys, impute missing variables, assign geography, construct the calibration targets, and calibrate. The calibration stage is the focus of this paper.

For this build the candidate universe is a three-year pooled ASEC support file for period 2024 (constructed as described below): 337,704 household records, one row per candidate household placement. We pin the exact build for reproducibility: POPULACE commit `558e46c1`, base H5 SHA-256 beginning `ec290055`, Ledger facts SHA-256 beginning `82cd9e87`, and target-registry version `5a22295b02b6`. The run resets household weights to a uniform prior before calibration, so the comparison is about support selection and reweighting rather than inherited survey weights.

7.2 Base microdata

7.2.1 Current Population Survey

The primary source is the Current Population Survey Annual Social and Economic Supplement (CPS ASEC; U.S. Census Bureau, 2024), a nationally representative household survey from the US Census Bureau and the Bureau of Labor Statistics. The CPS provides demographic structure, household relationships, labour income, transfer income, and reported participation in programmes such as SNAP, Medicaid, SSI, and TANF.

The candidate universe is not a single survey year. To widen the support before pruning, POPULACE pools the three most recently published CPS ASEC files—one per survey year—and ages the two earlier vintages forward to the latest year, so all three are expressed at a common period, 2024 for this build. Pooling three years rather than one substantially increases the number of candidate records and widens the variation the frame carries, while aging keeps the older vintages comparable to the most recent one; the result is a larger, richer candidate frame that still inherits the CPS survey design. This is the generate side of the generate-then-prune setting: the pooled file is deliberately larger than the deployable dataset, and calibration decides which of its 337,704 records survive.

The CPS is not a complete tax-benefit dataset on its own. Some tax variables are missing, others are measured with error, and top incomes are imperfectly captured relative to tax records (Burkhauser et al., 2012). Benefit receipt is also underreported relative to administrative sources (Meyer et al., 2015). The pipeline therefore augments the CPS before calibration.

7.2.2 Imputation from administrative and survey sources

The pipeline enriches the CPS base by imputing variables from administrative and survey donors. It imputes tax-return detail from the IRS Public Use File (PUF; Bryant, 2023) onto CPS records following Woodruff and Ghenis (2024a): the PUF carries detailed tax-return information but lacks household structure and demographic coverage, so quantile regression forests (Meinshausen, 2006) predict the PUF variables on the CPS; Meyer et al. (2020) assess the accuracy of such survey–administrative tax imputations. The imputation is sequential and regime-gated, drawing from a weighted bootstrap, so later variables condition on earlier imputed values. The same machinery draws further variables from the donors in Table 8 where the CPS is thin. Each step adds variables to the existing households rather than new records, so the candidate universe stays at one row per household.

Donor source	Variables contributed (this build)
CPS ASEC	Household and person structure, demographics, labour and transfer income, programme participation, unit assignment
IRS PUF (2015, uprated)	Itemized-deduction detail, QBI/SSTB components, partnership and S-corp income, selected tax detail
CPS ORG	Hourly wage, paid-hourly and union status, weekly hours, overtime inputs
SCF	Wealth and net-worth components
SIPP	Tips
MEPS-IC	Employer-sponsored insurance premiums
ACS (2022)	Rent (pre-subsidy)
CMS marketplace PUFs and CPS ASEC	ACA take-up and the benchmark-plan ratio

Table 8: Donor sources imputed onto the CPS base for this build, from the POPULACE-US release stage manifest. The stages run in a fixed order, each conditioning on the variables already imputed.

7.2.3 Geography

Each household carries a geographic identity. The CPS ASEC reports the household's state, and POPULACE attaches congressional-district identifiers so the same frame can be calibrated against national, state, and district targets. The current support file intentionally contains many more candidate household placements than the final deployable file should retain. That is the build-big, then-prune setting: POPULACE creates a support rich enough to cover the target surface, and the calibration method decides which records survive.

7.3 Calibration targets

The target system combines aggregates from several administrative sources at the national, state, and congressional-district levels. The headline experiment uses the full materialized target surface compiled from the current POPULACE US fiscal registry: 37,053 Ledger facts produce 32,633 targets after materialization against the candidate frame. Appendix A lists the target families and sources. The exact counts come from the run registry rather than from a pipeline-wide inventory.

Target family	Geographic level	Targets
IRS Statistics of Income (`irs_soi`)	National, state, district	31,350
Census population estimates (`census_pep`)	National, state	936
Medicaid and CHIP, CMS (`cms_medicaid`)	National, state	102
ACA marketplace, CMS (`cms_aca`)	State	102
SNAP, USDA (`usda_snap`)	National, state	52
State tax collections, Census STC (`census_stc`)	State	44
TANF, HHS ACF (`hhs_acf_tanf`)	National, state	30
Social Security supplement, SSA (`ssa_supplement`)	National	6
JCT tax expenditures (`jct`)	National	5
CBO revenue projections (`cbo`)	National	5
Medicare, CMS (`cms_medicare`)	National	1
Total		**32,633**

Table 9: Full materialized Populace target surface used in the matched full-surface probe, by family and geographic level. The surface contains 24,340 congressional-district targets, 7,815 state targets, and 478 national targets.

Counts are read from the run registry. In the headline experiment all targets, including `cbo`, are fit and scored; holdout variants are supplemental diagnostics (Section 8.5). The source for each family is listed in Appendix A.

8 Stage two: target-informed sparse selection

This section describes the calibration method that selects and weights records, and then the design of the experiment that evaluates the method as a sampler.

8.1 The shared calibrator

The calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$ maps n candidate records to m targets. Entry M_{ji} is the contribution of candidate i to target j: an indicator or count for count targets, and a simulated tax or benefit value for dollar targets. The matrix is built from PolicyEngine simulations evaluated under the policy rules that apply to each record's assigned geography, and is stored in sparse form. Given a weight vector $\mathbf{w} \in \mathbb{R}^n$, the estimate of target j is

$$\hat{t}_j = \sum_{i=1}^{n} M_{ji}\, w_i. \tag{8}$$

The methods compared in this study fit weights with the same calibrator, the `calibrate` routine in POPULACE. It minimizes a capped, weighted mean absolute percentage error (capped weighted MAPE) of the estimates against the targets,

$$\mathcal{L}_{\text{cal}}(\mathbf{w}) = \frac{\sum_{j=1}^{m} \omega_j \min\left(\left|\dfrac{\hat{t}_j - t_j}{s_j}\right|, c\right)}{\sum_{j=1}^{m} \omega_j}, \tag{9}$$

where $s_j = \max(|t_j|, 1)$ scales each target by its own magnitude, with a floor of one unit so a zero-valued target stays well defined; c caps the contribution of any single hard-to-fit target, so one badly conditioned target cannot dominate the gradient, and is set to the production value $c = 1$ in the runs reported here; and ω_j are per-target weights, normalised by their sum so only their relative magnitudes matter. The error is relative, so a one per cent miss on a small target and on a large one count the same. The weights are parameterized in log space, $w_i = \exp(\theta_i)$, so the fitted weights stay positive; an Adam optimizer (Kingma and Ba, 2015) implemented in PyTorch (Paszke et al., 2019) minimizes the loss, and the total weight is held to the input population. On a fixed set of records this calibrator answers a calibration question: it finds weights that match the targets. Dense no-L_0, post-L_0 refit, and random + reweight use it unchanged on different supports; informed L_0 extends it with a selection term.

The per-target weights ω_j in the runs reported here follow POPULACE's production US-fiscal weighting, computed through the same code path the production US build uses rather than a separate reimplementation. The weighting is built in four steps. Each target is first assigned to one of two value bases from its measure metadata: a count basis (indicator, enrollment, recipient, and return-count measures) or an amount basis (dollar measures). Within a basis, a target's raw weight is proportional to $\max(|t_j|, 1)^{1/2}$, the square root of its own magnitude, so a larger aggregate carries more weight but sublinearly (a target a hundred times larger receives about ten times the weight), and these raw weights are normalised to mean one inside each basis. The two bases are then rescaled to contribute equal total weight, so the far more numerous dollar cells do not swamp the count targets. A final step sets the mean weight to one. Because the loss divides by $\sum_j \omega_j$, only the relative weights enter the objective and their absolute scale does not.

8.2 Selection and weighting as one optimization

Informed L_0 sampling adds two terms to the shared calibrator that turn weight fitting into record selection. Each candidate i carries a Hard Concrete gate $z_i \in [0, 1]$ (Louizos et al., 2018), the gated estimate becomes $\hat{t}_j = \sum_i M_{ji} w_i z_i$, and the optimizer minimizes

$$\mathcal{L}(\mathbf{w}, \boldsymbol{\alpha}) = \mathcal{L}_{\text{cal}}(\mathbf{w}, \mathbf{z}) + \lambda_{\text{share}} \frac{1}{n} \sum_{i=1}^{n} \bar{z}_i + \lambda_{L_2} \frac{1}{n} \sum_{i=1}^{n} \left(\frac{w_i}{w_{0,i}} \right)^2 . \tag{10}$$

The first term is the capped weighted MAPE of Equation 9, now evaluated on the gated weights $w_i z_i$. The second term is the L_0 penalty and sets the sample size: $\bar{z}_i = \Pr(z_i \neq 0)$ is the expected activation of gate i (Equation 5), so $\sum_i \bar{z}_i$ is the expected number of open gates, the expected retained-record count, and λ_{L_0} sets how aggressively the optimizer prunes. In the full-surface probe we report λ_{L_0} through a normalized penalty share, $\lambda_{\text{share}} \sum_i \bar{z}_i / n$, so the sparsity price is comparable when the candidate record count changes. The raw coefficient passed to the optimizer is therefore $\lambda_{L_0} = \lambda_{\text{share}}/n$. The third term is a soft concentration penalty: λ_{L_2} multiplies the mean squared ratio of each fitted weight to its initial weight $w_{0,i}$, which discourages the optimizer from carrying the fit on a few heavily inflated records. It applies to informed L_0 only; the dense and random baselines run the shared calibrator without any selection or concentration term.

Read as a sampler, the calibration term keeps the retained sample faithful to the targets, the L_0 penalty sets the sample size, and the L_2 penalty governs weight concentration. The concentration term matters because an L_0 penalty on its own can reach a sparse solution that is still unusable: the optimizer can match the targets by placing very large weights on a few retained records. Both concentration controls act on the ratio of each fitted weight to its initial weight, $w_i / w_{0,i}$. The L_2 term is a soft penalty on the mean square of that ratio that discourages large inflations smoothly, and it is the natural control for tracing the accuracy-concentration trade-off. `max_weight_ratio` is a hard per-record cap on the same ratio: no fitted weight may exceed a fixed multiple of that record's initial weight, clamped after every optimizer step. The soft penalty shapes the whole weight distribution during optimization; the cap bounds the single largest weight. Size and concentration are therefore set by separate controls: λ_{L_0} sets how many records remain, and the L_2 penalty and `max_weight_ratio` can fix how concentrated their weights may become. The headline full-surface probe leaves the cap off and sets $\lambda_{L_2} = 0$, so the reported accuracy comparison is the pure-calibration result. Tightening either control is a natural robustness panel: it bounds how much population a single record can carry, at some cost to calibration fit.

Appendix B lists the full configuration, and Appendix C gives the optimization loop as Algorithm 1.

The gates and the weights are learned together: the gate decides whether a record is in the sample, and the weight decides how much it counts. Because both are fit against the same calibration loss, selection is informed by the targets: a record survives when keeping it helps match a target that would otherwise be missed. This is the sense in which the method is a target-informed sampler rather than a reweighting of a fixed random draw.

8.3 From fitted weights to a dataset

After optimization the gates are evaluated deterministically, so the retained-record count is read from the solution rather than chosen by a post-hoc cutoff. The retained records and their weights are assembled into a dataset for POLICYENGINE. Appendix D summarizes the assembly step, which matters operationally but is not the object of the experiment.

8.4 Sampling experiment design

The experiment evaluates whether informed selection produces a better dataset than the alternative samplers at the same record budget. The literature offers more possible comparators than can be run cleanly on the full mixed target surface: raking-based workflows are natural on categorical margins, balanced sampling is designed for fixed balancing variables, and combinatorial optimisation is discrete and costly at production scale. The headline comparison therefore keeps the candidate universe, target surface, positivity restriction, and calibration loss fixed, and varies only the way records are selected and, for the post-selection arms, refit. From one candidate universe and one target set, the matched full-surface probe compares five method arms: informed L_0 selection, informed L_0 selection followed by an ordinary calibration refit on the selected records, dense no-L_0 calibration on the full candidate universe, random sampling followed by gradient reweighting, and a no-refit random sample of dense calibrated weights scaled back to total mass. The run arms are:

- **Informed L_0 sampling with Hard Concrete gates.** The shared calibrator with the gate and concentration terms above selects records and fits their weights jointly at a chosen budget, set through λ_{L_0}.
- **Informed L_0 + refit.** The same selected records are kept, but the gates are removed and the shared calibrator refits ordinary dense weights on that retained subset, ini-

tialized from the informed L_0 weights. This post-L_0 refit isolates the value of the selected subset from the particular gated weights returned by the joint optimization.

- **Dense no-L_0 calibration.** The shared calibrator fits positive weights on the full 337,704-record candidate universe, with gates off and no sparsity or concentration penalty. This is the finite-optimization dense benchmark.

- **Random sampling with gradient-descent reweighting.** A random subset of the candidate universe of the same size is drawn, then the shared calibrator fits its weights to the same targets without the gate or concentration terms. This is the reduce-first, calibrate-after baseline.

- **Dense random scaled sample.** Dense no-L_0 weights are fit once on the full universe; a uniform random subset of the matched size is retained; the retained dense weights are then scaled so their total equals the dense total, with no post-sampling refit. This tests whether dense calibration alone can be made sparse by random thinning and mass scaling.

This design does not claim that raking, GREG, balanced sampling, or combinatorial optimisation are irrelevant in general. They remain the reference methods for simpler margin surfaces, classical calibration, target-balanced survey designs, and discrete synthetic-population construction. The empirical comparison below focuses on the methods that can be applied to the full Populace target surface under the same scoring loss and that were run in the real-data probe.

Holding the candidate universe and the target set fixed isolates the effect of how records are chosen. Every method starts from the same candidate frame with its weights reset to a uniform prior, so each record's initial weight $w_{0,i}$ is one shared constant. The reported probe uses $\lambda_{\text{share}} = 0.8$, which equals raw $\lambda_{L_0} = 2.37 \times 10^{-6}$ for the 337,704 candidate records and retains 57,240 records. The post-L_0 refit, dense no-L_0, and random + reweight baselines each use a 1,500-epoch optimizer budget; the sparse baselines use the same 57,240-record count. Sweeps across penalties and concentration controls are the next robustness panel rather than part of the headline evidence here.

8.5 Full-surface scoring

The production motivation is to construct a deployable dataset that fits the full target surface POPULACE currently assembles. The headline experiment therefore fits and scores every materialized target, including validation-only families and congressional-district targets. In the run reported here, 37,053 Ledger facts compile to 32,633 materialized calibration targets on 337,704 candidate household records. The reported objective is the same capped, weighted calibration loss in Equation 9, evaluated on the final retained weights and excluding the L_0 and L_2 penalties. This is the frontier an analyst would use when choosing a publishable record budget for the actual POPULACE build.

Generalization to structurally different targets is a separate question this proof of concept does not test. A meaningful holdout here would withhold whole target families rather than random target rows, because targets nest within a family (a national total is the sum of its state cells) and a random split would leak, with a held-out cell nearly determined by its retained siblings; we leave that family-level holdout robustness to future work (Section 10).

8.6 Evaluation metrics

Each run reports the metrics in Table 10 for the scored targets. They fall into two groups: accuracy measures, which ask how closely the weighted estimates reproduce the targets, and representativeness measures, which ask at what cost in weight concentration and compute that accuracy is bought.

Metric	Definition	Why reported
Populace objective loss	Equation 9: capped, target-weighted relative error, penalty-free	Headline target-fit metric; this is the loss the production calibrator optimizes
Median ARE	Median absolute relative error $\lvert\hat{t}_j - t_j\rvert / \lvert t_j\rvert$ over scored targets	Supplemental accuracy diagnostic, robust to the near-zero-denominator tail
Mean and max ARE	Mean and maximum of the same per-target errors	Supplemental, tail-sensitive diagnostics
Effective sample size	$\mathrm{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ and the implied design effect	A sampler can fit the targets while concentrating population mass on a few records, so ESS is a primary result, not a footnote
Retained count	Non-zero record count	Sample size
Max weight	Largest fitted weight	Weight concentration
Runtime	Wall-clock time	Compute cost

Table 10: Metrics reported for each run. Accuracy is led by the Populace objective loss, with raw ARE metrics reported as supplemental diagnostics; effective sample size measures the weight concentration that accuracy is achieved at.

Errors are reported by geographic level as well as in aggregate, because a sampler can match national totals while missing local ones.

A small number of targets have denominators so small that a single record can swing their relative error past 100 per cent. We identify these structurally rather than by an arbitrary cutoff: target j is denominator-degenerate when its value is smaller than its identifiability floor $\max_i \lvert M_{ji}\rvert\, w_{0,i}$, the largest contribution any single record makes at its initial weight. For such a target the absolute relative error reflects integer placement noise rather than calibration quality. Rather than winsorize or trim the error distribution, we name these targets in a one-time audit and report a targeted-removal sensitivity (the mean recomputed with exactly those named targets dropped) beside the full mean, so the effect of the degenerate targets is visible and attributable. Target counts are read from each run's manifest.

9 Calibration results

The results answer the production question directly: with the candidate universe, target surface, and record budget fixed, which retained weight vector best fits the current POPULACE target surface? All methods are scored on the full set of 32,633 materialized targets, using the penalty-free Populace objective loss in Equation 9. The run reported here is a matched full-surface probe rather than a full budget frontier: a normalized L_0 penalty selects a sparse support, the same retained count is used for the post-L_0 refit and sparse baselines, and an epoch-matched dense no-L_0 calibration anchors the comparison. Raw absolute-relative-error (ARE) summaries, effective sample size, and max weight are reported as diagnostics around the headline loss.

9.1 The full-surface matched probe

Figure 3 is the headline result. On the current three-year ASEC support file, the normalized L_0 penalty selects 57,240 of 337,704 candidate households, or 17.0% of the candidate universe. The raw gated L_0 weights are not the right publication weights: scored after thresholding, they reach a Populace objective loss of 9.86%. Removing the gates and refitting ordinary calibration weights on exactly the selected records lowers that loss to 4.74%. A dense no-L_0 calibration using all 337,704 records and the same 1,500-epoch optimizer budget reaches 5.07%. A random subset of the same size as the L_0 support, followed by the same 1,500-epoch gradient reweighting, reaches 7.55%. A random subset of the dense calibrated weights, scaled back to the dense total but not refit, reaches 24.24%.

The practical interpretation is that the L_0 selector is doing useful support-selection work, but the post-L_0 refit is essential. The selected support gives a lower finite-budget optimizer loss than the full dense run at 1,500 epochs and fits the full target surface much more closely than a random support of the same size. This should not be read as a claim that the dense feasible set is worse: the dense no-L_0 loss is still declining at the end of the 1,500-epoch budget. The defensible claim is that this L_0 support gives a better sparse support and a faster route to low loss under the completed optimizer budget. The current evidence should still be read as a single fixed-penalty probe, not as proof that this exact penalty is optimal: the next sweep needs to trace the normalized penalty across multiple retained counts and concentration settings.

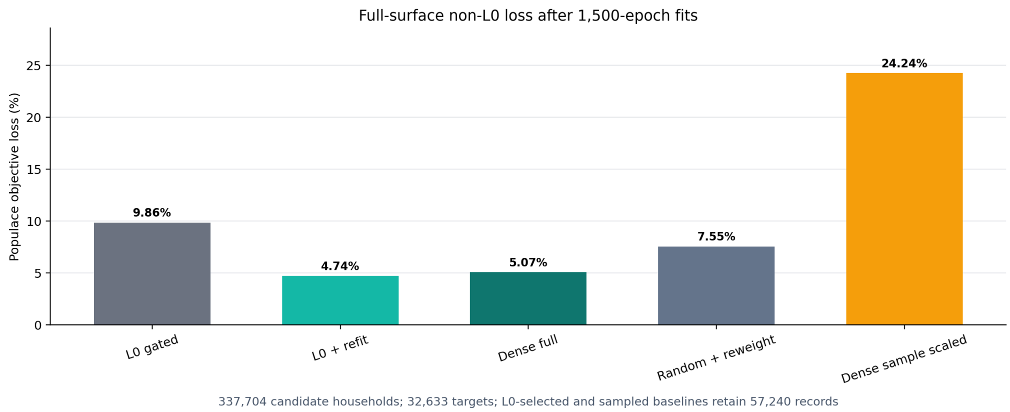

Figure 3: Full-surface Populace objective loss in the matched probe. The objective is the capped, target-weighted calibration loss in Equation 9, evaluated on each method's final retained/sampled weight vector and excluding sparsity or concentration penalties. The L_0 run uses a normalized penalty share of 0.8, equivalent to raw $\lambda_{L_0} = 2.37 \times 10^{-6}$ for 337,704 candidate households.

Table 11 reports the same comparison with raw ARE and weight diagnostics. The median ARE gives a different view from the capped objective: dense no-L_0 has the lowest median ARE (0.56%), while the post-L_0 refit is close (0.89%) and has the lowest capped Populace loss. Random + reweight has 6.70% median ARE, raw gated L_0 has 11.66%, and the dense random scaled sample has 55.24%. The mean ARE remains much larger because a small tail of difficult IRS targets can have very large relative misses even when the capped Populace loss is low.

Method	Records	Populace loss (%)	Median ARE (%)	Mean ARE (%)	ESS	Max weight
Informed L_0 + refit	57,240	4.74	0.89	84.64	4,726	913,836
Dense no-L_0 calibration	337,704	5.07	0.56	119.25	5,970	579,298
Random + reweight	57,240	7.55	6.70	113.88	2,480	1,163,939
Informed L_0 gated weights	57,240	9.86	11.66	89.84	6,879	546,455
Dense random scaled sample	57,240	24.24	55.24	436.13	981	2,176,121

Table 11: Full-surface comparison after 1,500-epoch fits. All rows use the same 337,704-record candidate universe, 32,633 targets, POPULACE production US-fiscal target-loss weights, and cap $c = 1$. "Dense random scaled sample" draws 57,240 records uniformly from the dense calibrated weights, keeps their dense weights, and rescales the retained weights to the dense total without refitting.

The L_0 run uses normalized penalty share 0.8, equivalent to raw $\lambda_{L_0} = 2.37 \times 10^{-6}$. Dense no-$L_0$ is still declining at epoch 1,500, so its row is an epoch-matched finite-optimization baseline rather than a convergence certificate.

Table 12 expresses the comparison on the headline loss scale. Relative to dense no-L_0, the post-L_0 refit lowers the completed-run Populace objective by 6.4%. Relative to random + reweight, it lowers the objective by 37.2%. Relative to the raw gated L_0 weights, it lowers the objective by 51.9%. The comparison isolates the value of the support and the refit: keeping dense weights on a random retained set and merely scaling them back to total mass performs poorly, so a sparse file still needs a reweighting step.

Comparison	Loss difference (pp)	Relative loss change
Informed L_0 + refit vs. dense no-L_0	-0.33	-6.4%
Informed L_0 + refit vs. random + reweight	-2.81	-37.2%
Informed L_0 + refit vs. raw gated L_0	-5.12	-51.9%
Dense random scaled sample vs. dense no-L_0	$+19.17$	$+378.3\%$

Table 12: Matched loss comparisons for the full-surface probe. Negative values favour the method named first. Differences are in percentage points of the capped, target-weighted Populace objective loss.

This table is descriptive because the current matched full-surface run uses one seed and one normalized L_0 penalty. Dense no-L_0, the post-L_0 refit, and random + reweight all use 1,500 optimizer epochs, but dense no-L_0 is still slowly declining at the end of that budget.

9.2 Size and concentration

The normalized penalty gives a usable size control in this probe. We parameterize the sparsity price as a share of the mean calibration loss, $\lambda_{\text{share}} \sum_i \bar{z}_i/n$, rather than as an unscaled raw coefficient. With $\lambda_{\text{share}} = 0.8$, the optimizer retained 57,240 records. This scaling matters operationally: the same raw coefficient has very different meanings when the candidate universe changes from tens of thousands to hundreds of thousands of records.

The selected support is more concentrated than the full dense calibration but better conditioned than the matched random sparse baselines. Dense no-L_0 has an effective sample size of 5,970 and a maximum weight of 579,298. The post-L_0 refit has ESS 4,726 and maximum weight 913,836. Random + reweight has lower ESS, 2,480, and a higher maximum weight, 1,163,939; the dense random scaled sample is worse again, with ESS 981 and maximum weight 2,176,121. The raw gated L_0 weights have higher ESS, 6,879, but worse target fit, which is why the refit is the publishable sparse variant.

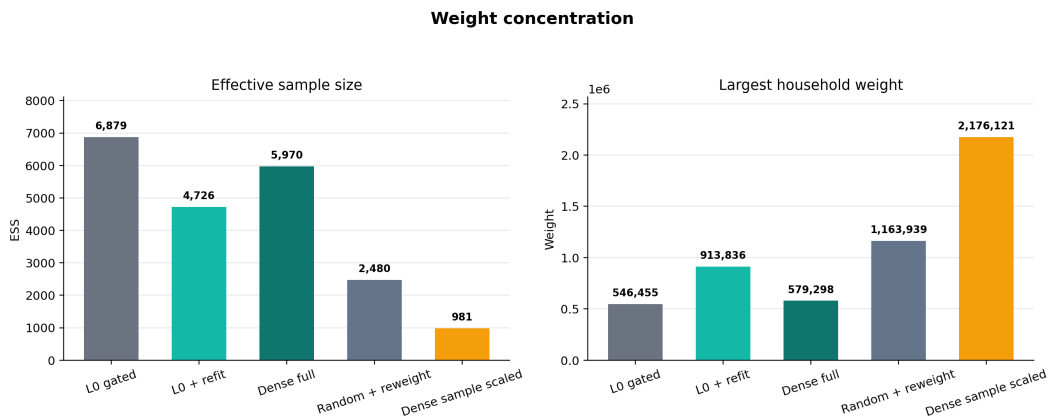

Figure 4: Weight concentration in the matched full-surface probe. Effective sample size and maximum weight are computed on each method's final retained/sampled weight vector. Dense no-L_0 retains all 337,704 records; the other sparse rows retain 57,240 records.

9.3 Accuracy by geography

Table 13 breaks the post-L_0 refit diagnostics down by geography. The fit is strongest at the state level, where median ARE is 0.31%, followed by national targets at 1.30% and district targets at 1.48%. District targets are the intended stress test for the method: they are the largest part of the target surface, and they are the reason POPULACE needs to build a large candidate universe before pruning it.

Geographic level	Targets	Median ARE (%)	Mean ARE (%)	Max ARE (%)
National	478	1.30	20.12	557.29
State	7,815	0.31	86.36	122,712.20
Congressional district	24,340	1.48	85.29	162,495.93
All scored targets	32,633	0.89	84.64	162,495.93

Table 13: Calibration accuracy by geographic level for the Informed L_0 + refit run at 57,240 records, measured as absolute relative error (ARE) across the full target surface. The median reports the typical target's error; the mean and maximum are tail-sensitive diagnostics.

Raw ARE is undefined for 26 IRS SOI national targets where both target and achieved values are zero; those rows are included in the target count and the Populace loss but omitted from the raw ARE summary.

9.4 Raw-error tails

The Populace objective is the headline score because it is capped and target weighted; raw ARE summaries are supplemental. In the matched probe, raw ARE is defined for 32,607 of the 32,633 scored targets. The remaining 26 rows are IRS SOI national targets with both target and achieved values equal to zero, so they are omitted from raw relative-error summaries but do not affect the capped loss. Among the defined rows, the post-L_0 refit has a 0.89% median ARE and an 84.64% mean ARE. The gap between the median and the mean is the reason the paper leads with the Populace loss and reports raw ARE only as a diagnostic.

9.5 Cost

The matched probe took 39.4 minutes for the L_0 selection run. The post-L_0 refit then added about 9.5 minutes, for a combined 48.9 minutes reported in the artifact. Dense no-L_0 took 38.7 minutes for 1,500 epochs, random + reweight took 7.3 minutes, and the dense random scaled sample required no extra optimization beyond the dense fit. The selector is therefore computationally expensive, and the dense run is still slowly declining at the endpoint. The accuracy gain over dense and random + reweight is large enough in this probe to justify a broader normalized-penalty sweep, but not enough to skip the next robustness work: more penalties, longer dense convergence checks, more candidate-support regimes, and concentration controls. Figure 5 in Appendix E plots wall-clock runtime against accuracy across requested record budgets as a supplemental cost diagnostic from the earlier budget sweep.

10 Future work

This study moves the proof of concept onto POPULACE's current full target surface, including congressional-district targets. The next empirical step is to turn the matched probe into a frontier by sweeping the normalized sparsity share across retained counts. Holdout panels should remain separate robustness checks that ask narrower questions: which target families or geographies generalize when withheld, which targets dominate the loss, and how sensitive the result is to the production target weights and cap.

A related thread varies the penalties the headline run deliberately holds fixed. The reported experiments set $\lambda_{L_2} = 0$ and leave the weight cap off, so the comparison isolates the effect of the L_0 gates on sparsification; this leaves the concentration controls as a clean axis to study on their own. Three experiments follow. First, evaluate the convex L_1 selector of Section 6.6 head-to-head with L_0 on the same surface: because a single L_1 penalty couples how many records survive with how much the survivors shrink, we expect it to be a weaker selector than the L_0 gates, and running it would show whether that coupling actually costs it. Second, sweep the soft concentration penalty λ_{L_2} and the hard `max_weight_ratio` cap against λ_{L_0}. Third, use that sweep to trace the effective-sample-size against accuracy tradeoff, quantifying how much calibration fit a build gives up to bound weight concentration.

A second thread is the build-large-then-prune regime the controls are designed for, in which a pipeline assembles a candidate universe far larger than it can ship—hundreds of thousands or millions of records rather than a compact survey file—and prunes it to a deployable budget under the calibration targets. The current three-year support file is a step in that direction. More aggressive support expansion is where target-informed selection has its clearest opportunity to justify its cost: a random draw from a tight budget is least likely to span a demanding target system, so a selector trained on the targets has the most to add.

The full-surface design also clarifies which classical comparators are worth adding. Survey-weight (probability-proportional-to-size) sampling is the natural calibrate-then-reduce counterpart to random + reweight and belongs in the next frontier as a literature comparator, with the caveat that it selects records by the population mass they represent rather than by their fit to the targets. Raking, GREG, balanced sampling, and combinatorial optimization remain relevant reference methods, but not all are natural baselines for a heterogeneous 32,633-target fiscal surface. The appropriate next comparison is on the simpler categorical-margin subsets where their assumptions hold cleanly, in the spirit of Tanton et al. (2014)'s comparison of generalized regression with combinatorial optimisation for small-area reweighting, while the production frontier continues to use the Populace loss over the complete surface.

11 Application: legislative incidence at the district level

The pipeline's output is in production use. The 57,240-record sparse support selected in section 9 ships, after post-selection refit, as the certified United States release of POPULACE, pinned by immutable release tag and consumed through `policyengine.py`'s managed data bundle alongside a pinned model version. A companion paper (Ghenis et al., 2026) uses it to decompose the One Big Beautiful Bill Act of 2025 — eighteen tax provisions and three program-participation scenarios — against a TCJA-expiration counterfactual, reporting incidence at the national, state, and congressional-district level, with poverty and inequality statistics computed on the same file.

Three aspects of that deployment bear directly on this paper's claims. First, the imputed inputs stage one exists to provide are load-bearing: the companion analysis's tip, overtime, auto-loan, and program-participation provisions run entirely on imputed or enhanced columns, and its health-inclusive resource measure depends on the modeled program dollars the enhanced file supports.

Second, the value of target-informed selection is directly observable in the district layer. The companion paper's district estimates use weights calibrated against roughly 24,000 district-level targets. Recomputing its district averages on assignment-only weights — the same national file, with records assigned to districts but with no district targets in the calibration — produces a minimum district cell of 34 records, a district with a negative average net-income change, and an interdecile district ratio of 3.0 against the calibrated layer's 2.2: tail artifacts of sparse, unbalanced support rather than incidence. Random assignment is to district estimation what random subsampling is to national calibration in section 9 — the untargeted baseline that target-informed optimization exists to beat.

Third, deployment makes data-stack sensitivity measurable. Rerunning the companion decomposition with identical provision dictionaries on the retired predecessor data stack moves the headline aggregate by $66 billion and flips the sign of the deep-poverty response, while headline signs — aggregate direction, a bottom-decile loss, a rising Gini alongside a falling top-1% share — agree across stacks. Incidence conclusions inherit the data pipeline; publishing the pipeline, its calibration diagnostics, and cross-stack comparisons alongside the policy numbers is what makes that inheritance auditable.

12 Discussion

Read together, the two stages yield an operational recipe. *Fill with the design*: fit imputation models weighted, gate zero-inflated components, and chain dependent targets — the failures of doing otherwise are invisible to headline joint-geometry metrics and surface only in the tails and in classifier two-sample tests, exactly where transfer-program and top-income analysis is most exposed. *Select with the targets*: treat reduction to a simulable support as estimation, let the calibration targets drive record retention, and refit publication weights on the selected support rather than publishing the gated weights that selected it. The population-view harness connects the stages: the same latent population, viewed through surveys for evaluating fill and through administrative aggregates for evaluating representation.

The limitations are those of each stage plus one that emerges at the seam. Stage one's results are single-imputation; imputation variance and multiple-imputation inference are future work, and the tail-extension question — how far beyond the donor support imputed tails should extend — is deliberately left open. Stage two's dense-calibration baseline was still improving at its optimization budget, so the sparse-versus-dense comparison is a finite-optimization result on a fixed support, not a convergence claim; concentration controls and normalized-penalty sweeps remain to be mapped. At the seam, certification currently gates the national and state target families while congressional-district targets ship as diagnostics — the assignment-only contrast in section 11 is the measured cost of that gap, and promoting district families into the certified gate is the pipeline's next milestone. Finally, the companion deployment shows that residual data defects propagate into named policy provisions (a tip-income carrier undercount halves a tip-exemption estimate); the discipline the pipeline enables is not defect-free data but disclosed, provision-level accounting of which defects bind where.

13 Conclusion

Constructing representative microdata for sub-national policy analysis is two estimation problems, not one: filling in what the base survey does not observe, and choosing — then weighting — the records that will represent the population at every geography a policymaker asks about. This paper presented both stages as they are implemented and deployed in POPULACE: a weighted, regime-gated, chained quantile-regression-forest imputation operator whose design choices are each attributed by ablation under a population-view harness, and a target-informed L_0 selection-and-calibration step whose 57,240-record support, refit after selection, outperforms both dense calibration and random subsampling on a 32,633-target administrative surface and ships as the certified United States release. A companion incidence analysis of the One Big Beautiful Bill Act consumes that release end to end, making the pipeline's choices — and its disclosed defects — auditable in a live legislative application at national, state, and district grain.

The near-term agenda follows directly: promote district target families into the certified gate; map the penalty–concentration frontier and extend the dense-calibration budget; add multiple-imputation variance and principled tail extension to the fill stage; and grow the population-view harness into a standing release gate, scoring every candidate build against every survey view and administrative surface before it ships. The estimator is installable independently of the stack as `populace-fit`, and every number in both stages reproduces from committed runs in the accompanying repositories.

Conflict of interest

All authors are affiliated with POLICYENGINE, the nonprofit organization that develops POPULACE, `populace-fit`, and `microimpute`, the software evaluated in this paper. The authors have no other competing interests.

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Data and code availability

The paper and experiment code are at https://github.com/PolicyEngine/imputation-paper; every results table regenerates from committed run configurations via the repository's command line. The estimator is implemented in `populace-fit`, part of POPULACE, open source at https://github.com/PolicyEngine/populace. Baseline implementations are `microimpute` (https://github.com/PolicyEngine/microimpute) and `py-statmatch` (https://github.com/CosilicoAI/py-statmatch). The reported sweeps ran against `populace-fit` at populace commit `71b5a83`, `microimpute` 1.1.2, and `py-statmatch` at commit `094a242`; the SCF 2022 summary extract from the Federal Reserve (SHA-256 beginning `3bb4d890`) and the Census ASEC 2025 public-use bundle (`asecpub25csv.zip`, SHA-256 beginning `318845a2`) are the pinned survey inputs — both read directly from their agencies of origin — and each run directory's manifest records rows, caps, seeds, and methods.

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A Calibration targets

This study calibrates to the full materialized target surface POPULACE currently constructs for the US fiscal build. Every target is a source-backed fact from PolicyEngine Ledger (PolicyEngine, 2026), the `arch-data` fact store that holds the source publications and preserves the provenance of each published value; Populace composes calibration targets from those facts. The surface combines national, state, and congressional-district aggregates across the major tax and transfer families (Table 14). It is reported here in full so the empirical claims can be read against the exact target system used rather than against a pipeline-wide inventory.

Table 14 lists each family with its administrative source. The system holds 32,633 targets across eleven families and three geographic levels: 24,340 congressional-district, 7,815 state, and 478 national targets. The `irs_soi` family supplies 31,350 of them (96%), so any error summary aggregated over all targets is dominated by IRS income-tax aggregates.

Family	Administrative source	Levels	Targets
irs_soi	IRS Statistics of Income: Historic Table 2 state totals and congressional-district tables	National, state, district	31,350
census_pep	Census Bureau annual state resident population estimates	National, state	936
cms_medicaid	CMS Medicaid and CHIP applications, eligibility, and enrollment reports	National, state	102
cms_aca	CMS open-enrollment-period state-level public use file	State	102
usda_snap	USDA FNS SNAP monthly state participation and benefit summaries	National, state	52
census_stc	Census Bureau State Tax Collections (STC), item T40 individual income taxes	State	44
hhs_acf_tanf	HHS ACF federal TANF and state MOE financial data	National, state	30
ssa_supplement	SSA Annual Statistical Supplement, OASDI beneficiaries and benefits	National	6
jct	JCT estimates of federal tax expenditures	National	5
cbo	CBO revenue projections by category	National	5
cms_medicare	CMS Medicare Trustees Report	National	1
Total			**32,633**

Table 14: Calibration target families and their administrative sources. Counts and sources are read from the full-surface target registry used by the run. In the headline experiment all listed families are fit and scored (Section 8.5).

B Optimization hyperparameters

Table 15 lists the hyperparameters of the L_0 optimization with their roles. The gate parameters follow the Hard Concrete construction of Louizos et al. (2018). The sparsity penalty λ_{L_0} and the epoch count are set per build to reach a target dataset size and are reported with each run.

Parameter	Value	Role
Gate temperature β	0.25	Sharpness of the 0/1 gate transition
Left stretch γ	-0.1	Enables exact-zero gates after clipping
Right stretch ζ	1.1	Enables exact-one gates after clipping
Initial keep probability	0.8	Records start mostly active while leaving room for the sparsity penalty to close gates
Weight jitter SD	0.05	Log-space noise on weights at initialization
Logit jitter SD	0.01	Log-space noise on gate logits at initialization
Learning rate	0.02	Adam optimizer step size
Epochs	1,500	Training iterations in the reported fixed-penalty probe
Target loss cap c	1	Caps a single target's contribution to $\mathcal{L}_{\text{cal}}$; the runs reported here use Populace's production US-fiscal value $c = 1$
λ_{share}	0.8	Normalized sparsity share in $\lambda_{\text{share}} \sum_i \bar{z}_i/n$; raw $\lambda_{L_0} = 2.37 \times 10^{-6}$ for 337,704 records
λ_{L_2}	0	Soft concentration penalty on $(w_i/w_{0,i})^2$ (mean over records); held at zero in the reported full-surface run
Budget bisection steps	None	The reported full-surface probe uses one fixed normalized penalty; a future frontier should sweep this value
`max_weight_ratio`	None (uncapped)	Hard per-record weight cap: $w_i \leq$ ratio $\times$ $w_{0,i}$, clamped each step

Table 15: Hyperparameters for the L_0 optimization. Gate parameters follow Louizos et al. (2018). Source: the `l0-python` Hard Concrete implementation used by POPULACE.

C Algorithm pseudocode

Algorithm 1 presents the L_0-regularized calibration procedure.

Algorithm 1 L_0-regularized calibration with Hard Concrete gates

Require: Calibration matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$, targets $\mathbf{t} \in \mathbb{R}^m$, initial weights $\mathbf{w}_0 \in \mathbb{R}^n$, per-target loss weights $\boldsymbol{\omega} \in \mathbb{R}^m$

Require: Hyperparameters: λ_{L_0}, λ_{L_2}, weight cap r (`max_weight_ratio`; ∞ if uncapped), β, γ, ζ, learning rate η, epochs E

Ensure: Calibrated sparse weight vector $\hat{\mathbf{w}} \in \mathbb{R}^n$

Initialize $\log w_i \leftarrow \log w_{0,i} + \mathcal{N}(0, 0.05^2)$ for all i

Initialize $\log \alpha_i \leftarrow \text{logit}(0.8) + \mathcal{N}(0, 0.01^2)$ for all i

Initialize Adam optimizer with parameters $\{\log w_i, \log \alpha_i\}$ and learning rate η

for epoch $= 1$ to E **do**

  Sample Hard Concrete gates (training):

  for $i = 1$ to n **do**

    $u_i \sim \text{Uniform}(\epsilon, 1 - \epsilon)$

    $s_i \leftarrow \sigma\left(\frac{\log u_i - \log(1-u_i) + \log \alpha_i}{\beta}\right)$

    $\bar{s}_i \leftarrow s_i(\zeta - \gamma) + \gamma$

    $z_i \leftarrow \min(1, \max(0, \bar{s}_i))$

  end for

  $w_i^{\text{eff}} \leftarrow \exp(\log w_i) \cdot z_i$ for all i

  $\hat{t}_j \leftarrow \sum_i M_{ji} \cdot w_i^{\text{eff}}$ for all j

  $\mathcal{L}_{\text{cal}} \leftarrow \dfrac{\sum_{j=1}^{m} \omega_j \, \min\left(\left|\frac{\hat{t}_j - t_j}{s_j}\right|, c\right)}{\sum_{j=1}^{m} \omega_j}$

  $\mathcal{L}_{L_0} \leftarrow \sum_{i=1}^{n} \sigma\left(\log \alpha_i - \beta \log \frac{-\gamma}{\zeta}\right)$

  $\mathcal{L}_{L_2} \leftarrow \frac{1}{n} \sum_{i=1}^{n} \left(\frac{\exp(\log w_i)}{w_{0,i}}\right)^2$ ▷ soft concentration; informed L_0 only

  $\mathcal{L} \leftarrow \mathcal{L}_{\text{cal}} + \lambda_{L_0} \mathcal{L}_{L_0} + \lambda_{L_2} \mathcal{L}_{L_2}$

  Backpropagate $\nabla_{\log w, \log \alpha} \mathcal{L}$

  Adam step

  Clamp $\log w_i \leftarrow \min(\log w_i,\ \log(r \cdot w_{0,i}))$ for all i ▷ per-record cap; no-op if $r = \infty$

end for

Deterministic inference:

for $i = 1$ to n **do**

  $z_i^{\text{det}} \leftarrow \min(1, \max(0,\ \sigma(\log \alpha_i)(\zeta - \gamma) + \gamma))$

  $\hat{w}_i \leftarrow \exp(\log w_i) \cdot z_i^{\text{det}}$

end for

return $\hat{\mathbf{w}}$

D Dataset assembly details

After optimization the fitted weight vector is mapped back to record form and used to assemble the deployable dataset. The assembly step keeps the active records, reconstructs their person and tax-unit memberships, derives geography from the stored assignment, and recomputes geography-dependent quantities. Two details matter for fidelity. Geography is derived from the same assignment used to build the calibration matrix, so the assembled dataset matches the universe the optimizer saw. Take-up draws are regenerated with the same deterministic seeds used during matrix construction, so take-up-dependent targets stay consistent between the calibration problem and the assembled dataset.

E Cost diagnostics

Figure 5 is retained as a supplemental cost diagnostic from the earlier budget sweep. The matched full-surface probe's current runtimes are reported in Section 9: dense no-L_0 took 38.7 minutes for 1,500 epochs, L_0 selection plus refit took 48.9 minutes combined, and random + reweight took 7.3 minutes.

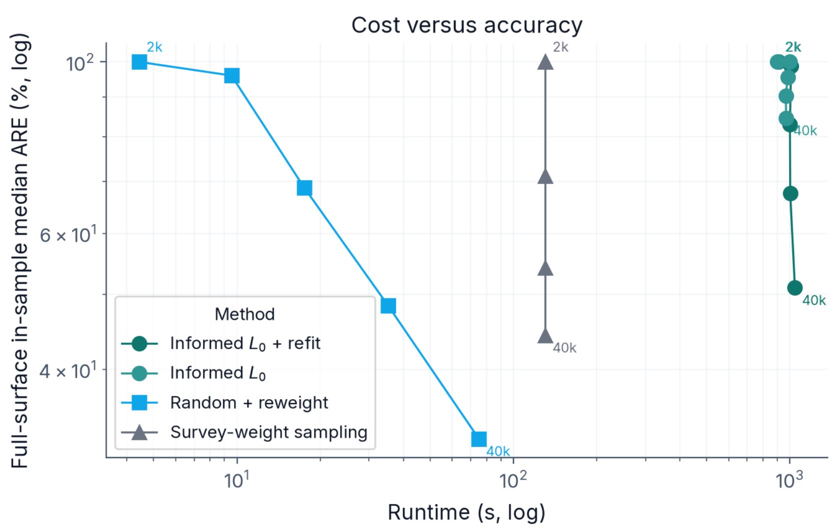

Figure 5: Supplemental historical cost against accuracy: wall-clock runtime (log scale) against full-surface median ARE (log scale), each point labelled by its requested record budget. This figure predates the fixed normalized-penalty full-surface probe and is used only to motivate reporting runtime alongside accuracy.